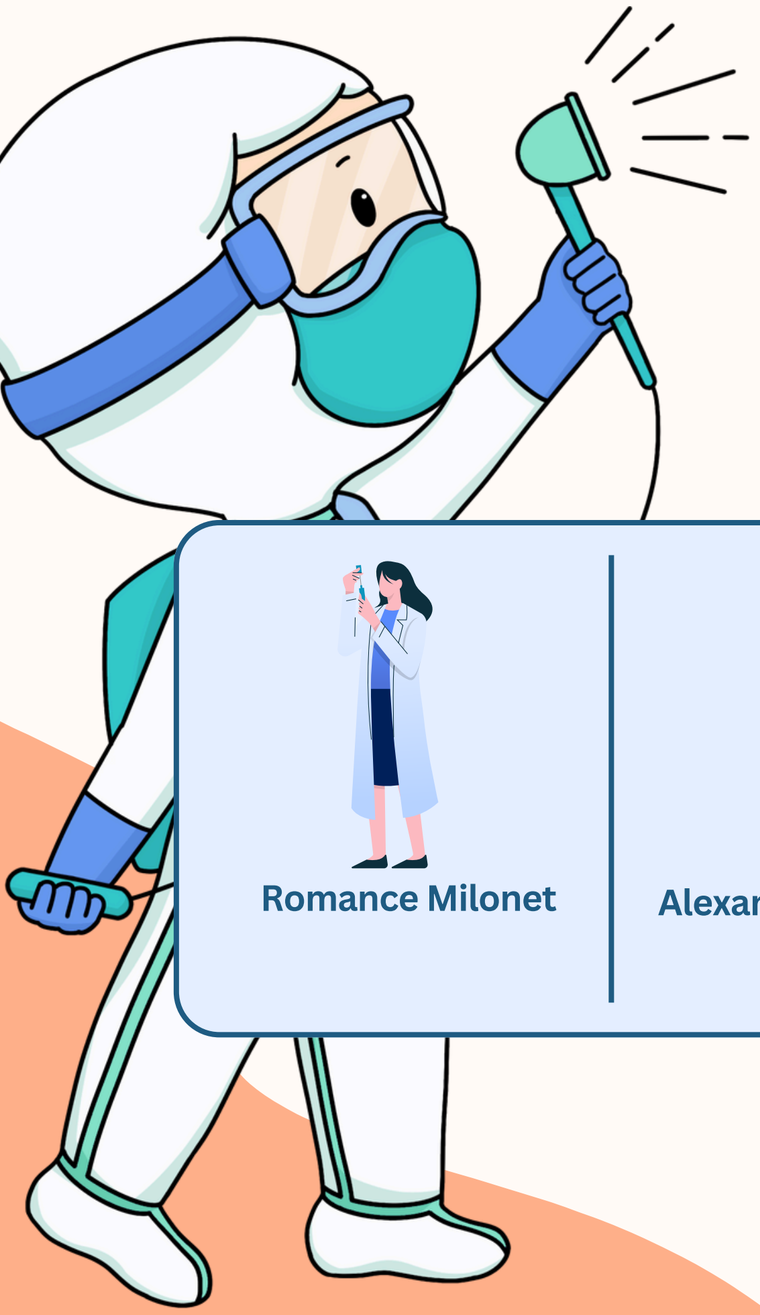
Institut Pasteur

Dengue Epidemic Study



Our team

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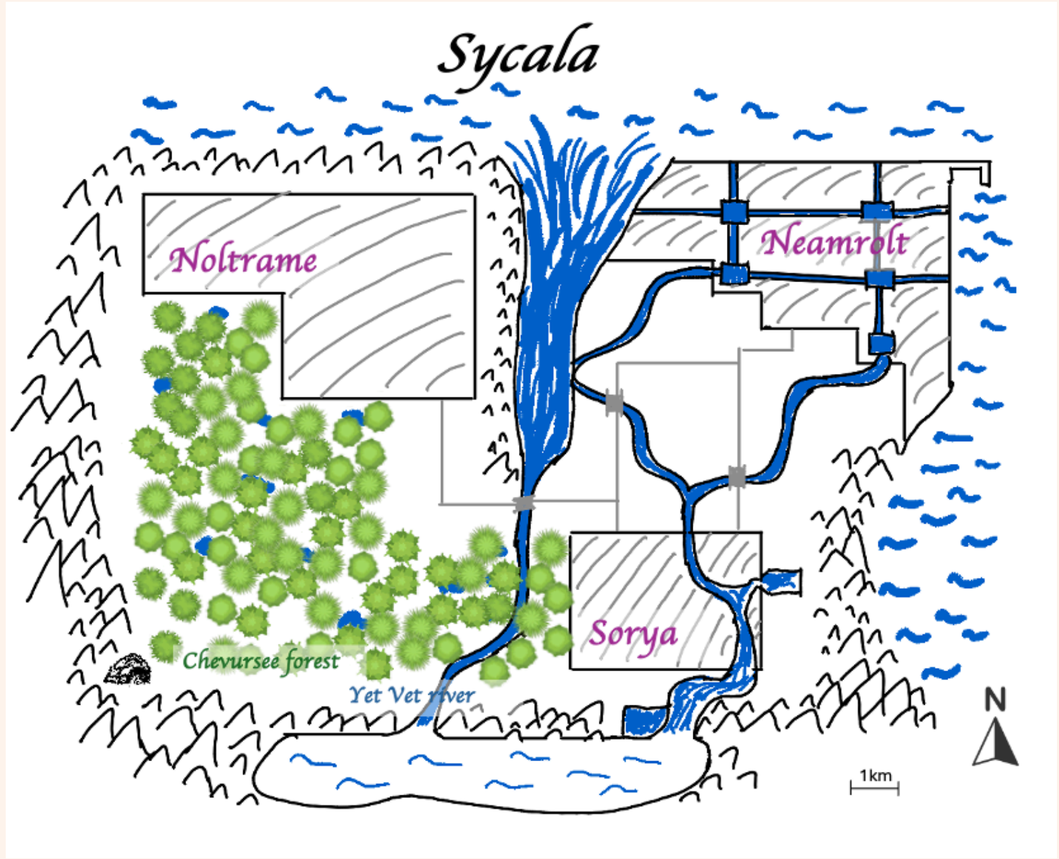
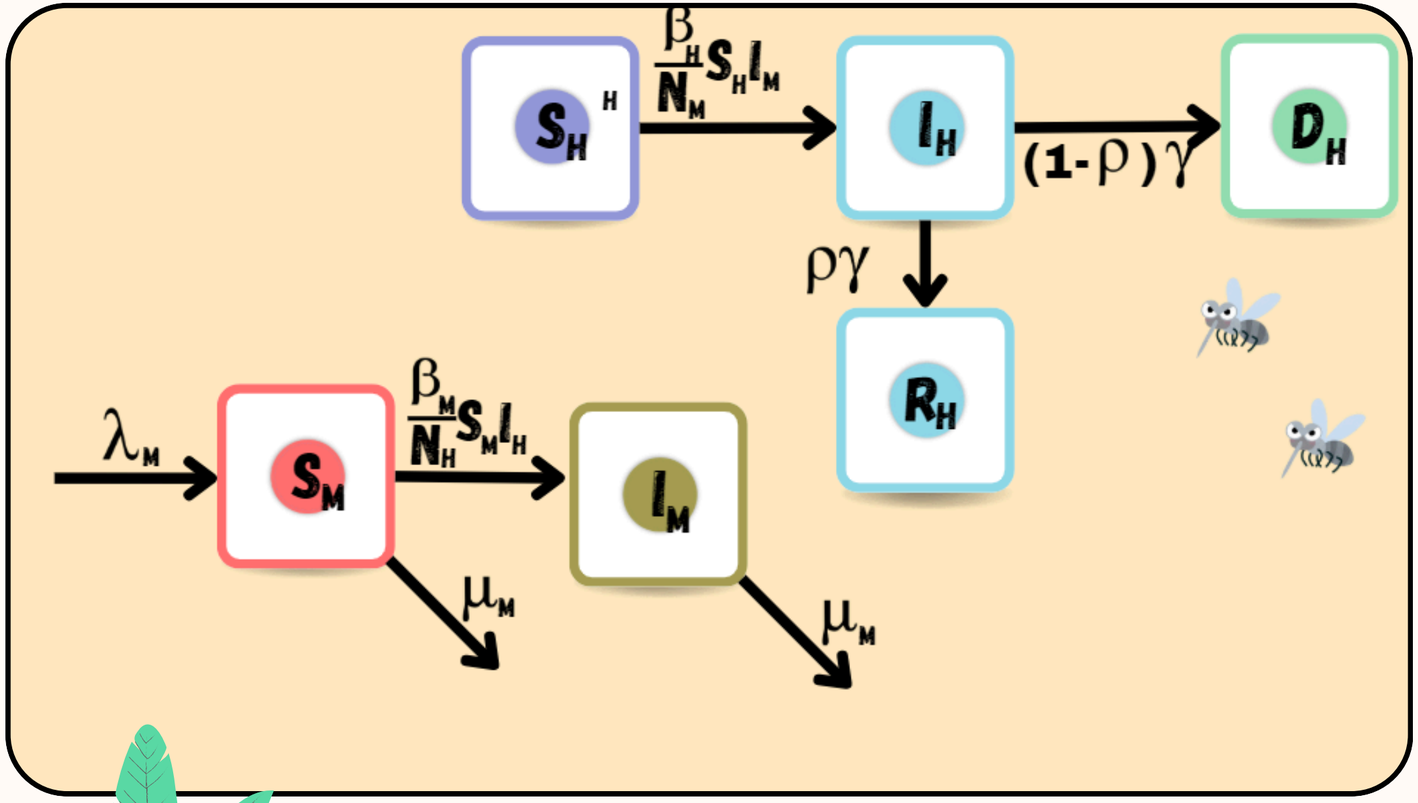
DENGUE IN SYCALA : A PUBLIC HEALTH CRISIS



THE BASE OF OUR MODELS



Sh , Sm : susceptible
Ih , Im : infected
Rh : recovered
Dh : dead / hospitalized
Nh , Nm : total population

ρ : probability of dying from the disease γ : recovery rate β : infectivity rate λ : mosquito birth rate μ : mosquito death rate

A simple model:
-SIRD
-SI

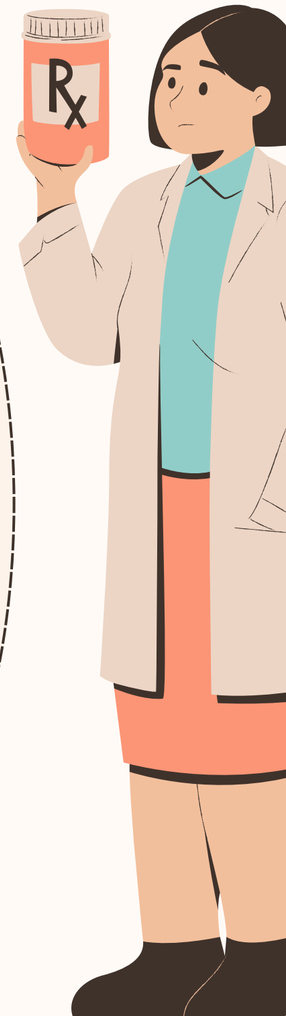
EQUATIONS FOR SIRD / SIS MODEL

Choice of :

$$\frac{\beta}{N}$$

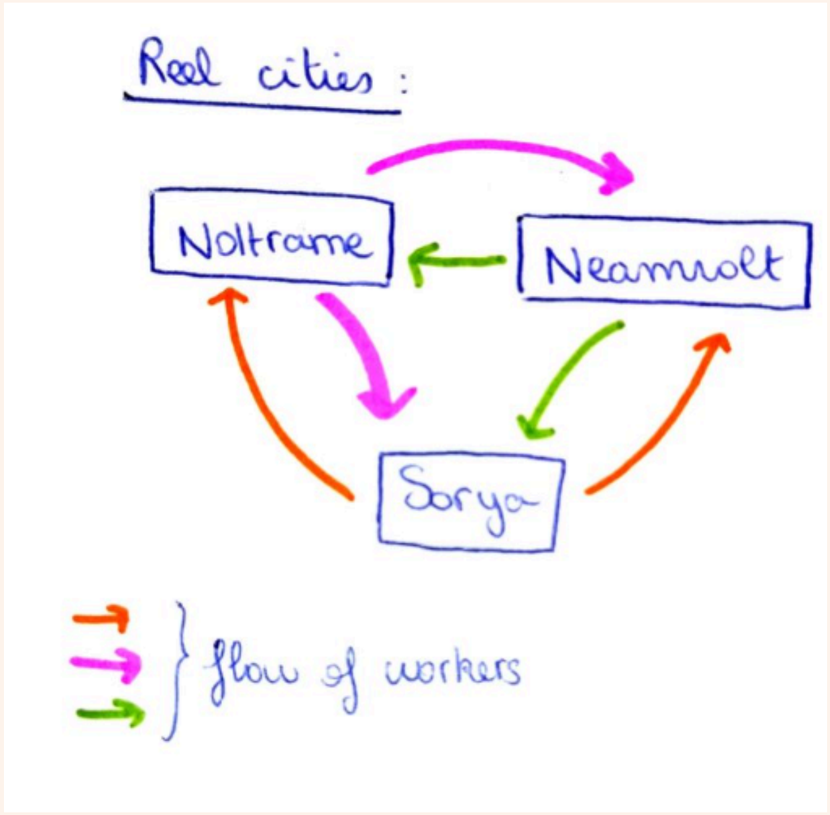
Disease transmission is frequency dependent because mosquitoes don't bite more if the population is bigger.

$$\frac{dS_h}{dt} = -\beta_h \frac{S_h I_m}{N_m}$$
$$\frac{dI_h}{dt} = \beta_h \frac{S_h I_m}{N_m} - \gamma_h I_h$$
$$\frac{dR_h}{dt} = (1-\rho)\gamma_h I_h$$
$$\frac{dD_h}{dt} = \rho_h I_h$$
$$\frac{dS_m}{dt} = \lambda N_m - \beta_m \frac{S_m I_h}{N_h} - \mu S_m$$
$$\frac{dI_m}{dt} = \beta_m \frac{S_m I_h}{N_h} - \mu I_m$$

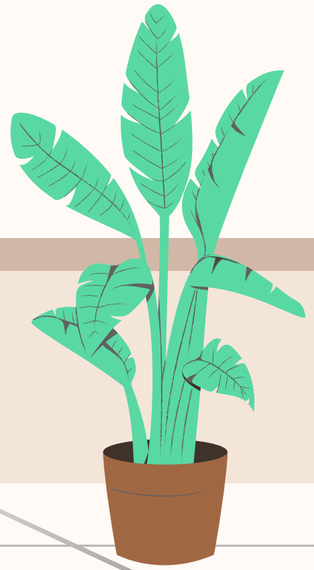


WORK FLOW BETWEEN CITIES

Overview 01



¨People work in other cities

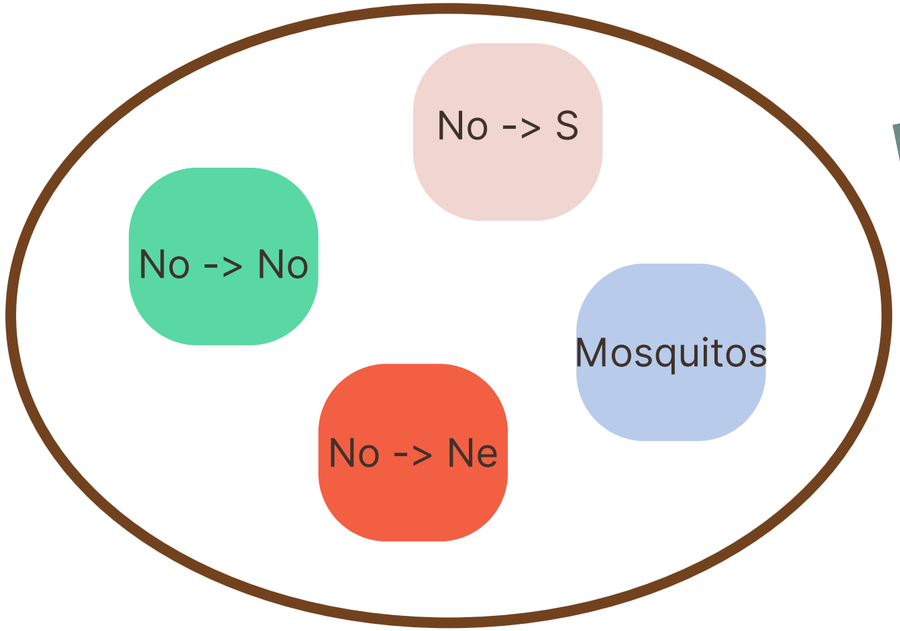


MODEL WITH WORK FLOW

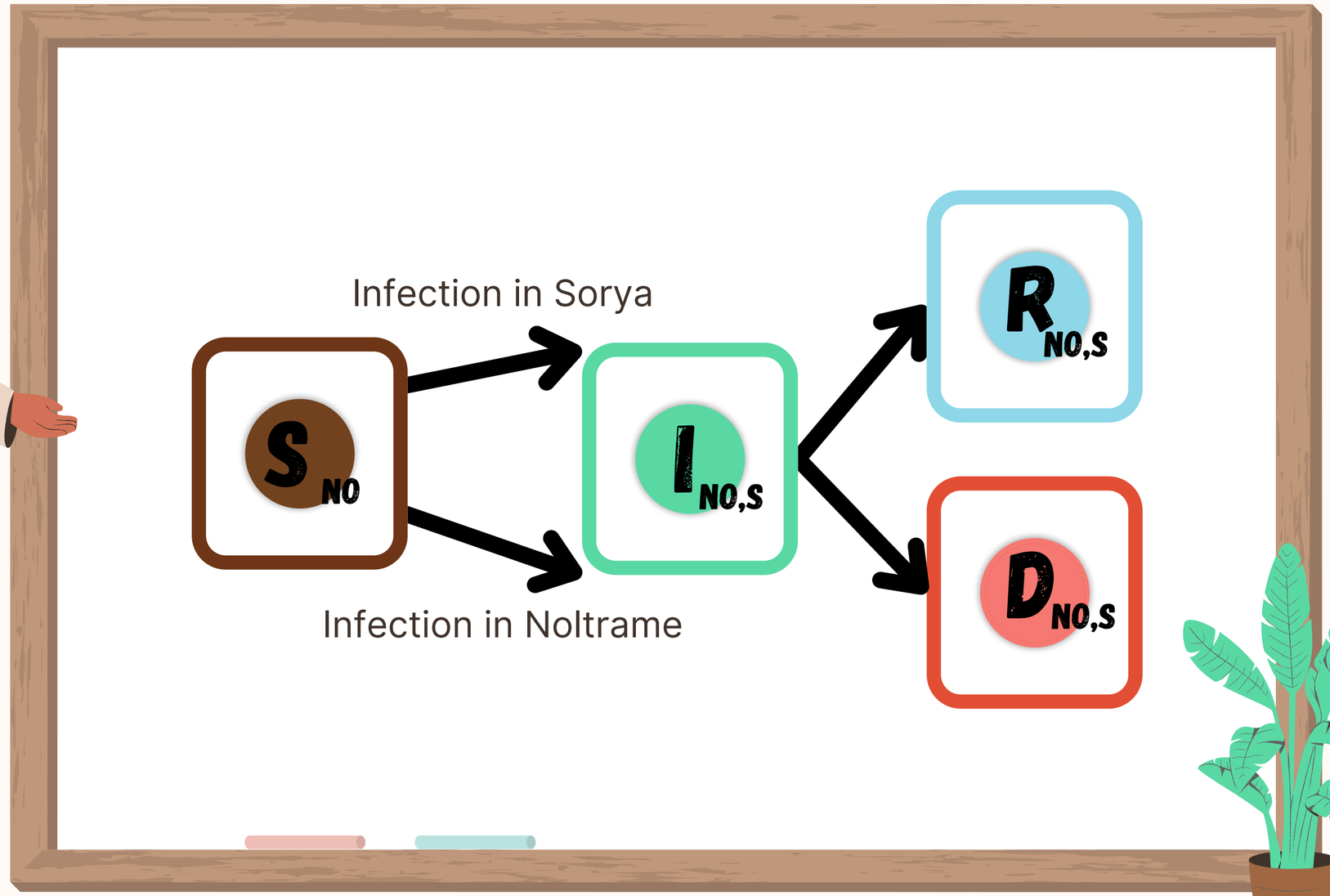
For each city : 4 populations

3 SIRD for humans for each city

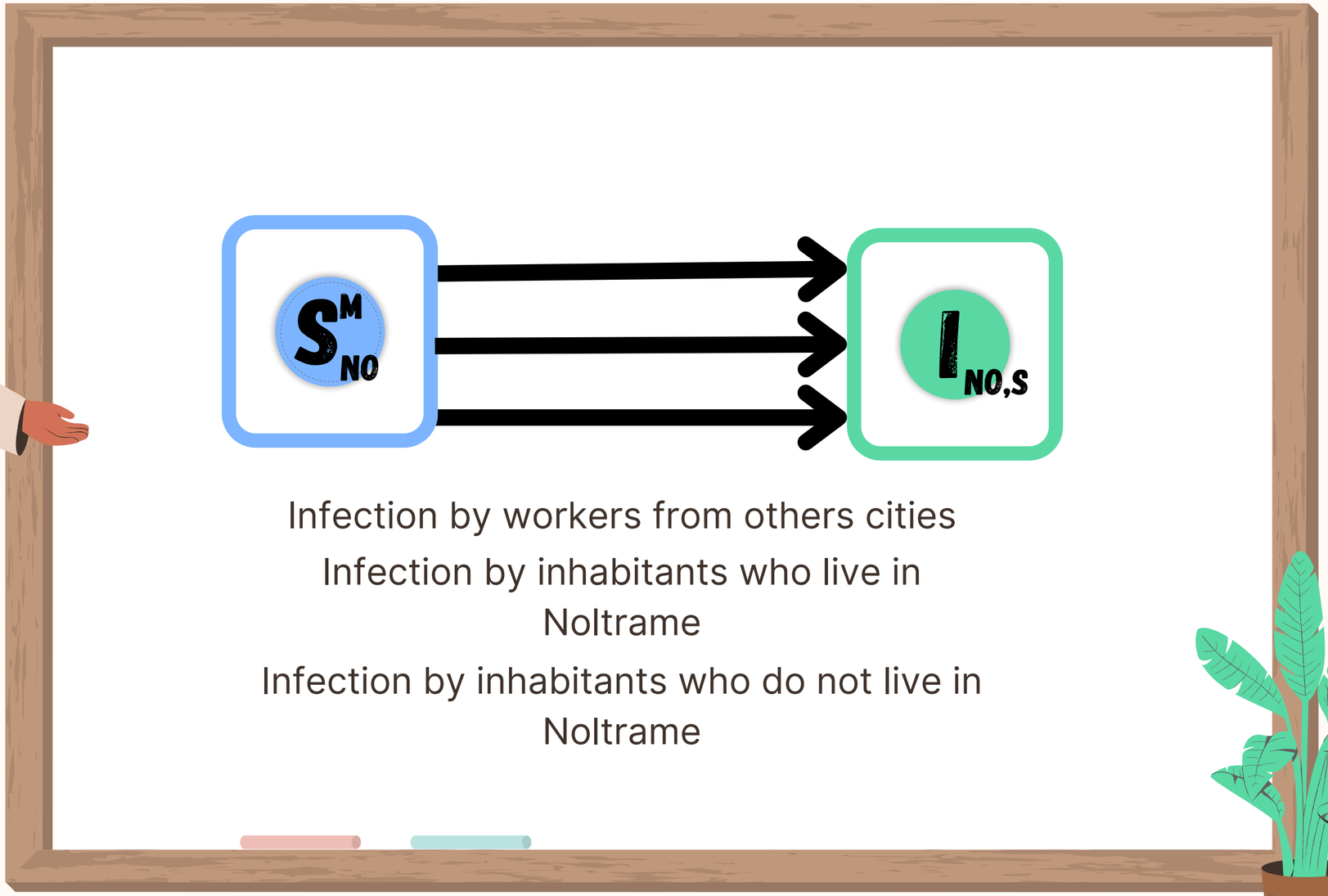
SI for mosquitoes for each city



MODEL WITH WORK FLOW



MODEL WITH WORK FLOW



Infection by workers from others cities

Infection by inhabitants who live in Noltrame

Infection by inhabitants who do not live in Noltrame

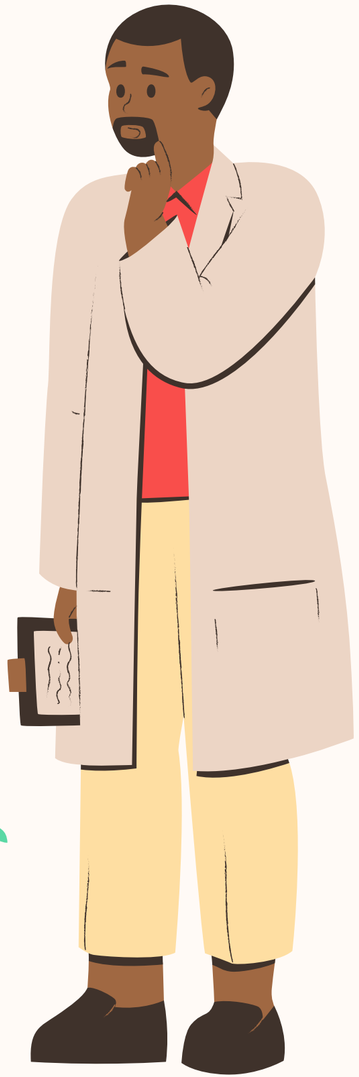
MODEL WITH WORK FLOW

Exemple for people of Noltrame working in Sorya :

$$\frac{dI_{n_o \to s}}{dt} = \frac{\beta_{nm}}{N_{nm,s}} I_{nm,s} \; S_{n_o \to s} \times \text{time-work}$$

$$+ \frac{\beta_{nm}}{N_{nm,n_o}} I_{nm,n_o} \; S_{n_o \to s} \times (1 - \text{time-work})$$

$$- \gamma I_{n_o \to s}$$



ASSUMPTIONS AND SIMPLIFICATIONS

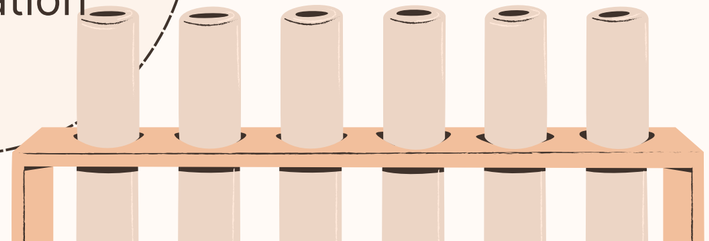
Discussing efforts to combat infectious diseases and pandemics

Assumptions

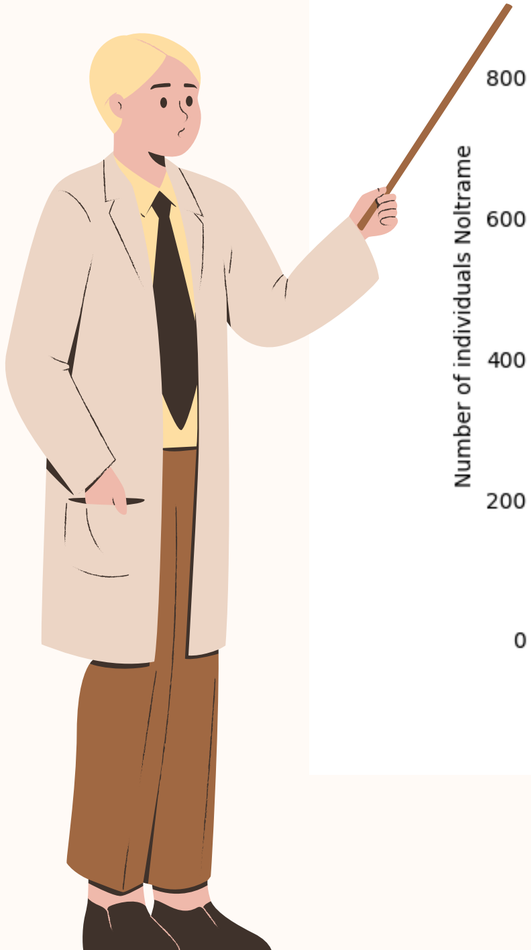
- same life-style during the pandemic
- same tendency to be bitten by a mosquitoe
- same chance to contract the disease
- Mosquitoes stay in the same city
- No mosquitoes in Neamrolt

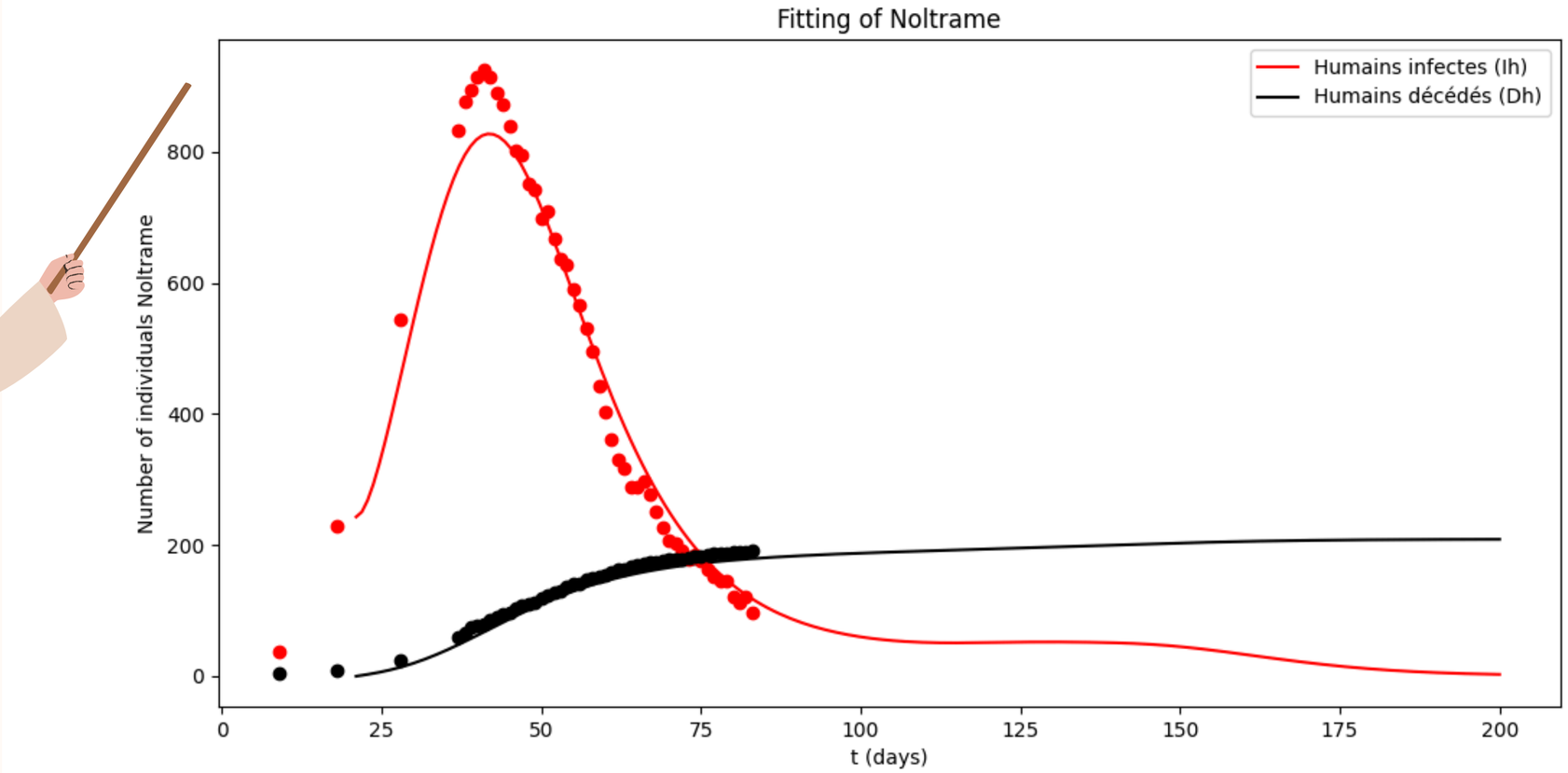
Simplifications

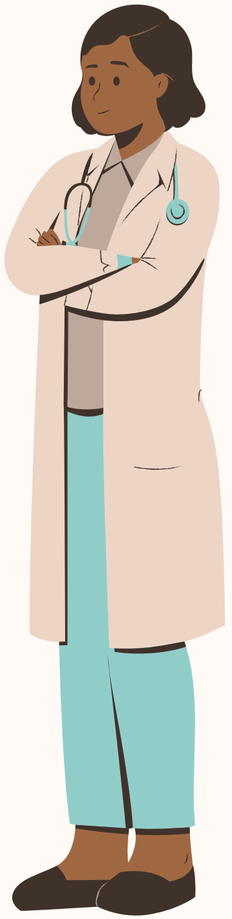
- We do not take into account the drive from one city to another
- We do not take into consideration the city's demographic



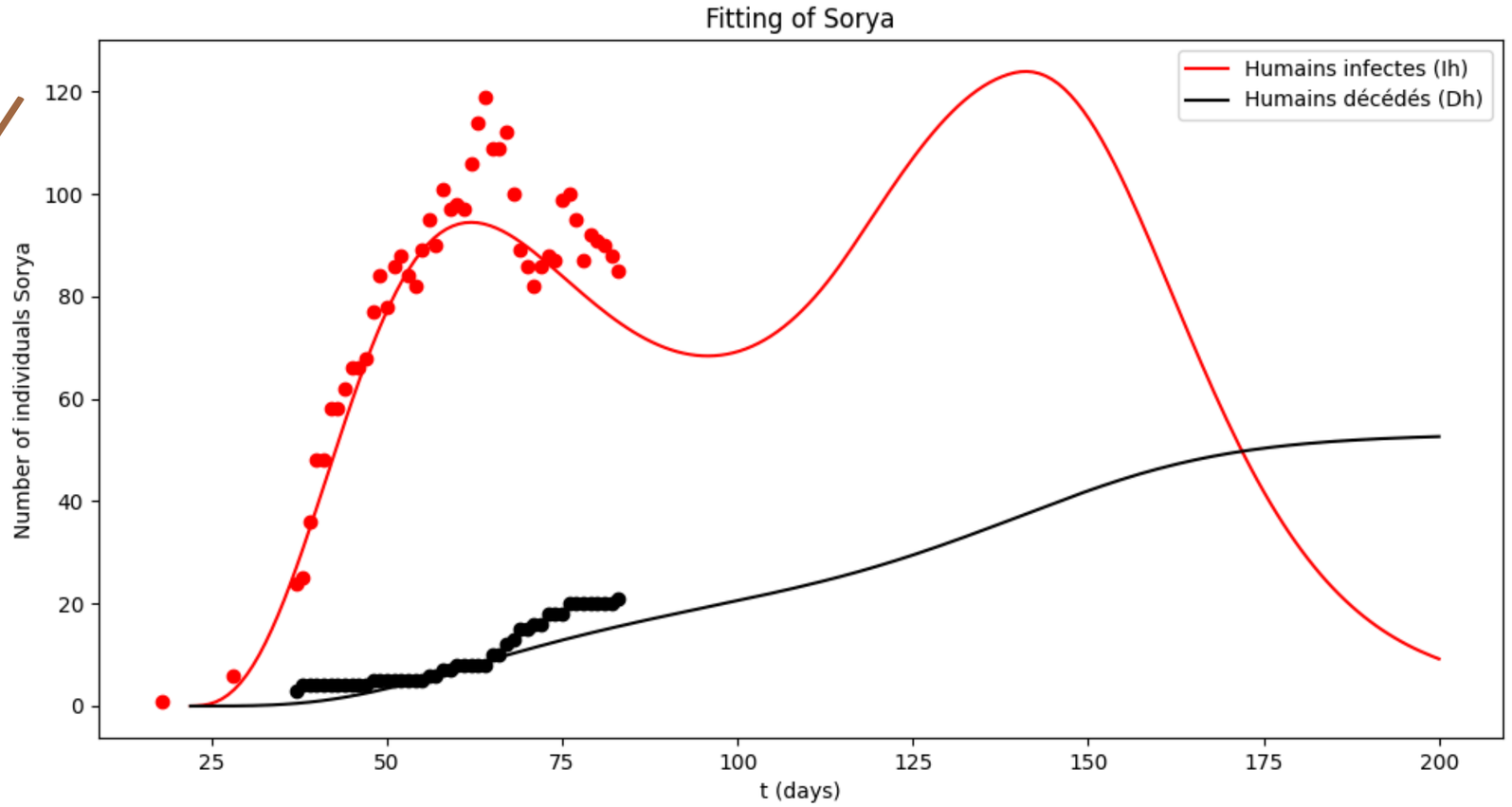
PREDICTIONS AND FITTING

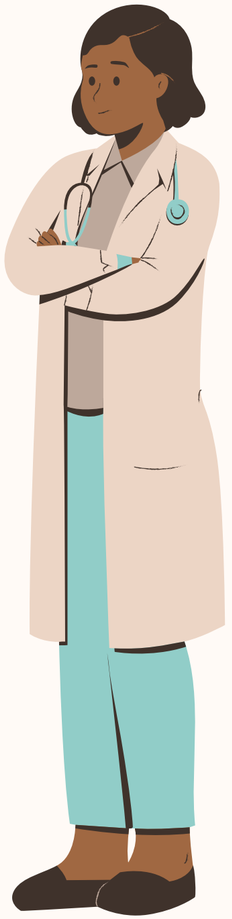




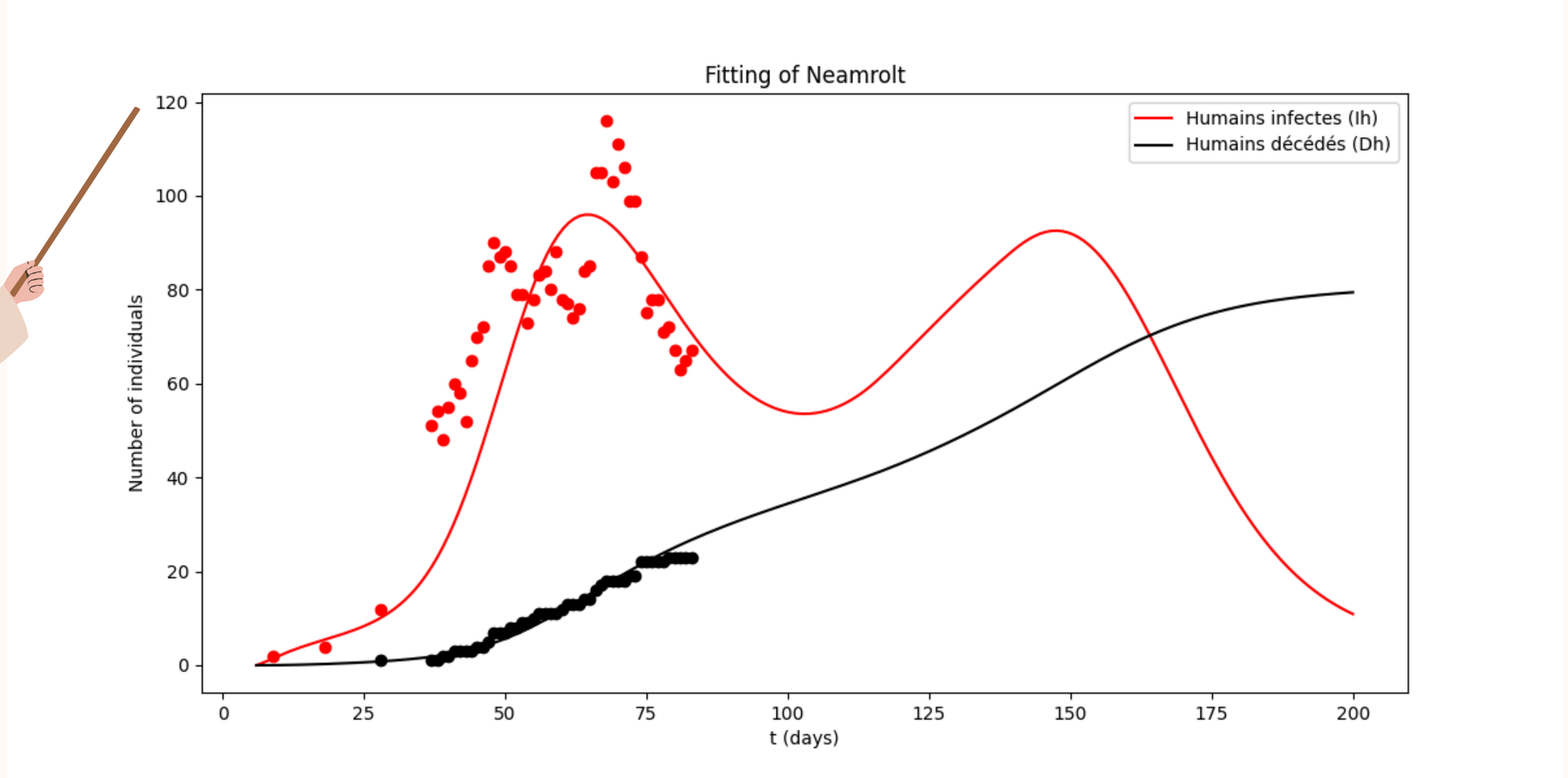


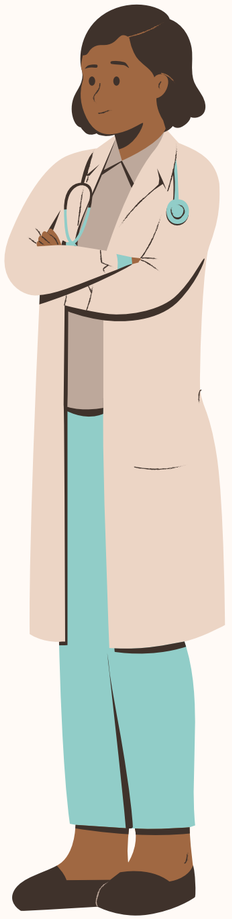
PREDICTIONS AND FITTING





PREDICTIONS AND FITTING





PREDICTIONS AND FITTING

Discussing the role of fitting

```
def error(params_to_fit, times, data):
    c_ne_fit ,gamma_h_ne_fit, rho_ne_fit, mu_m_fit, p_ne_fit, pm_ne_fit = params_to_fit

    # Un fitting peut etre fait sur un nb arbitraires de variables. Les variables sans "_fit" restent FIXES
    # et ne seront pas fitees. On fite tres rarement sur mu parce que ça donne des resultats aberrants
    # par rapport a la population de moustiques
    params = (c_ne_fit ,gamma_h_ne_fit, rho_ne_fit, mu_m, p_ne, pm_ne_fit)

    # On resout l'equa diff
    model_predictions = integrate.odeint(deriv, Y0["no"], times, args=params)

    # On calcule les erreurs
    diff = np.array([])
    diff = np.append(diff, (data[0] - model_predictions[:, 1]))
    diff = np.append(diff, (data[1] - model_predictions[:, 2]))
    diff = np.append(diff, (data[2] - model_predictions[:, 3]))

    # On precise qu'il faut bien ignorer les cas ou c'est pas defini
    diff[np.isnan(diff)] = 0
    return (diff)

#Utilisation de la methode des moindres carres pour determiner les coeffs
fitted_params = optimize.least_squares(
    error,
    param_values_tab["no"],
    bounds=([0, 0, 0, 0, 0, 0],[np.inf, 1, 1, 1, 1, 1]),
    args=(time, [I_observed, D_observed, R_observed]))
```

PREDICTIONS AND FITTING

Discussing the role of fitting

$$\theta = \begin{pmatrix} c \\ \gamma_h \\ \rho \\ p_h \\ p_m \end{pmatrix}$$

$$\varepsilon(\theta) = \sum_{i \in \mathrm{Data}} \left(y_\theta(i) - y_{\mathrm{obs}}(i)\right)^2$$

$$\theta_{\mathrm{opt}} = \arg \min_{\theta \in \mathbb{R}_+ \times [0,1]^4} \varepsilon(\theta)$$

PREDICTIONS AND FITTING

Discussing the role of fitting

Parameter	Physical interpretation	Constant / Fitted	Why?	Value
μ	Mosquito death rate	Constant	Studies have analysed the life cycle of mosquitoes	1/28
β_h	Transmission rate from mosquitoes to humans	Fitted	We only have an approximation of the rate	0.3
β_m	Transmission rate from humans to mosquitoes	Fitted	We only have an approximation of the rate	1
P and T	Ratio of time : Work / Home	Constant	We supposed people work 40% of their time	0.4 and 0.6
γ	Recovery rate	Fitted	Unknown precise value	0.2
ρ	Infection death rate	Fitted	Unknown precise value	0.03

EXTRACTING DATA

Habitants de Neamrolt :

- 6573 travaillent à Sorya
- 416 travaillent à Noltrame

Habitants de Sorya :

- 487 travaillent à Neamrolt
- 137 travaillent à Noltrame

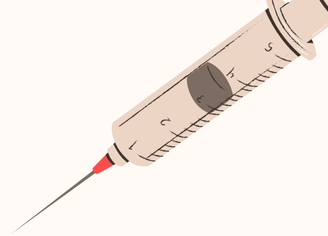
Habitants de Noltrame :

- 5571 travaillent à Sorya
- 1091 travaillent à Neamrolt

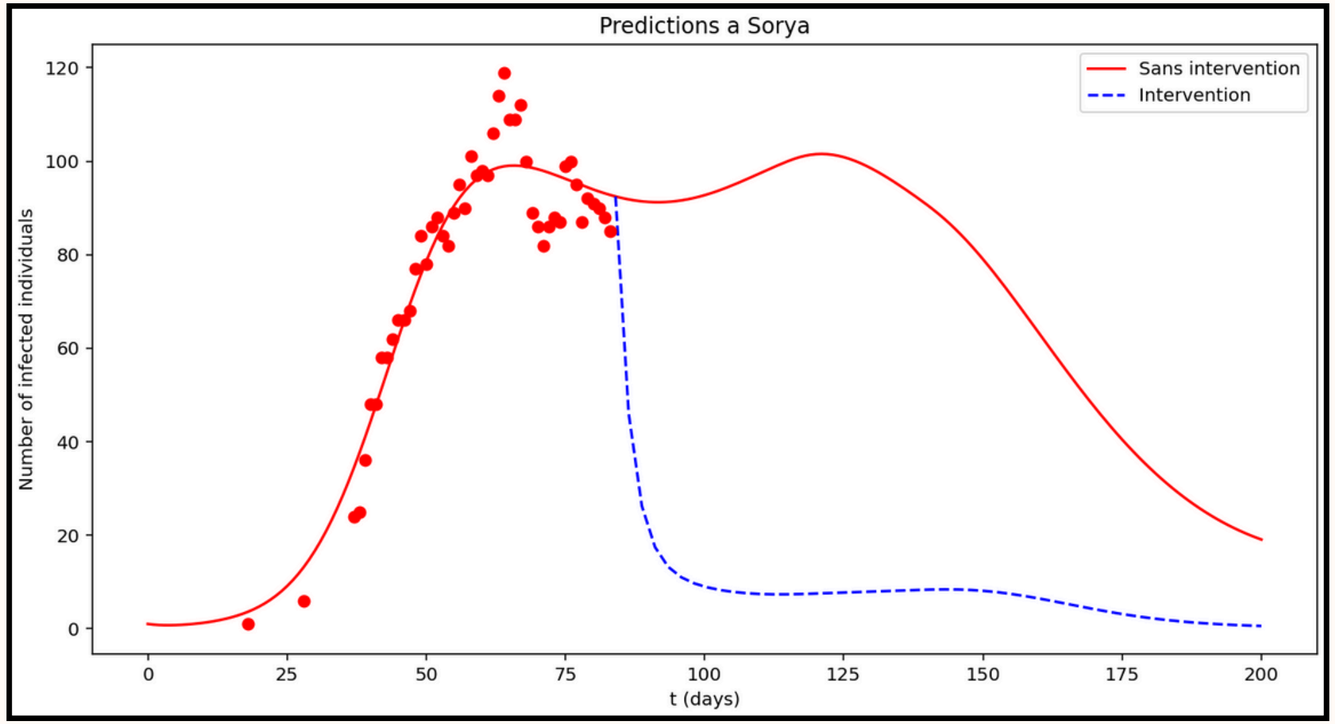
```
1   Select company, count(*) from citizens group by company
```

	company	count(*)
1	APS	2112
2	EDE	1494
3	McLilg	849
4	NDA	1500
5	OCM	1451
6	OFC	1481
7	PSNe	540
8	PSNo	891
9	PSS	1389
10	SGI	1396
11	SHP	2246
12	ST	1435
13	STCo	1516

OUR INTERVENTION PROPOSITION



Synthesis of the expenses				
Intervention	Noltrame	Neamrolt	Sorya	Sycala
Confinment	$ -	$ -	$ -	$ -
Chomage	$ -	$ -	$ -	$ -
Larvicide	$ -	$ -	$ -	$ -
Insecticide	$ -	$ -	$ -	$ -
Spray	$ 48 750,00	$ 15 000,00	$ 44 700,00	$ 108 450,00
Sterile Males	$ -	$ -	$ -	$ -
Mosquito Nets	$ -	$ -	$ -	$ -
Vaccine	$ -	$ -	$ -	$ -
Treatment	$ 70 000,00	$ 160 000,00	$ 60 000,00	$ 290 000,00
All	$ 118 750,00	$ 175 000,00	$ 104 700,00	$ 398 450,00
Authorized Budget:	Noltrame	Neamrolt	Sorya	Sycala
				$ 400 000,00
				$1 550,00

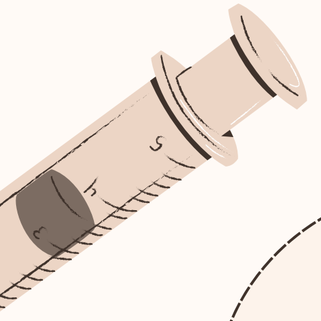


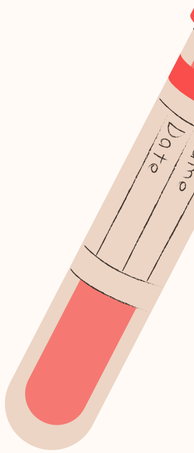
- Spray
- Standard treatment

Cost : $398 450

SCIENTIFIC PERSPECTIVES

IMPROVING EPIDEMIC MODELING FOR FUTURE DENGUE OUTBREAKS





Overview 01

Start modeling with simple hypotheses

Always integrate real data-streams

Overview 03

Gradually take more aspects into account in your model (seasonality, population movements, localization, etc.)

Overview 02

Collaborate and discuss with everyone you can, so that you have all of the information you need

Conclusion

Dengue remains a global health crisis

While the epidemic in Sycala is under controle, future outbreaks in other regions are possible.

Data matters

Real-time data collection significantly improves model reliability.

Modeling is key

A well-thought-out model is essential for predicting the evolution of the epidemic and indentifying the best interventions to mitigate it.

Collaboration is essential

International cooperation, interdisciplinary research and open data sharing are key to mitigating an epidemic.

KEEP FIGHTING

Modeling is hard but you will ALWAYS learn from it, so never give up and do the best you can to save lives.

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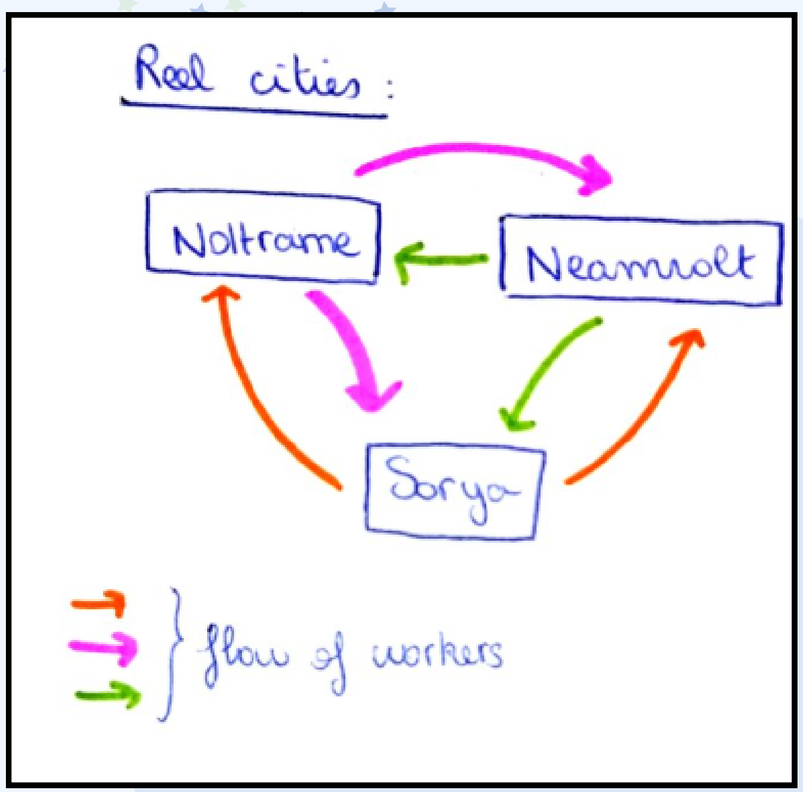
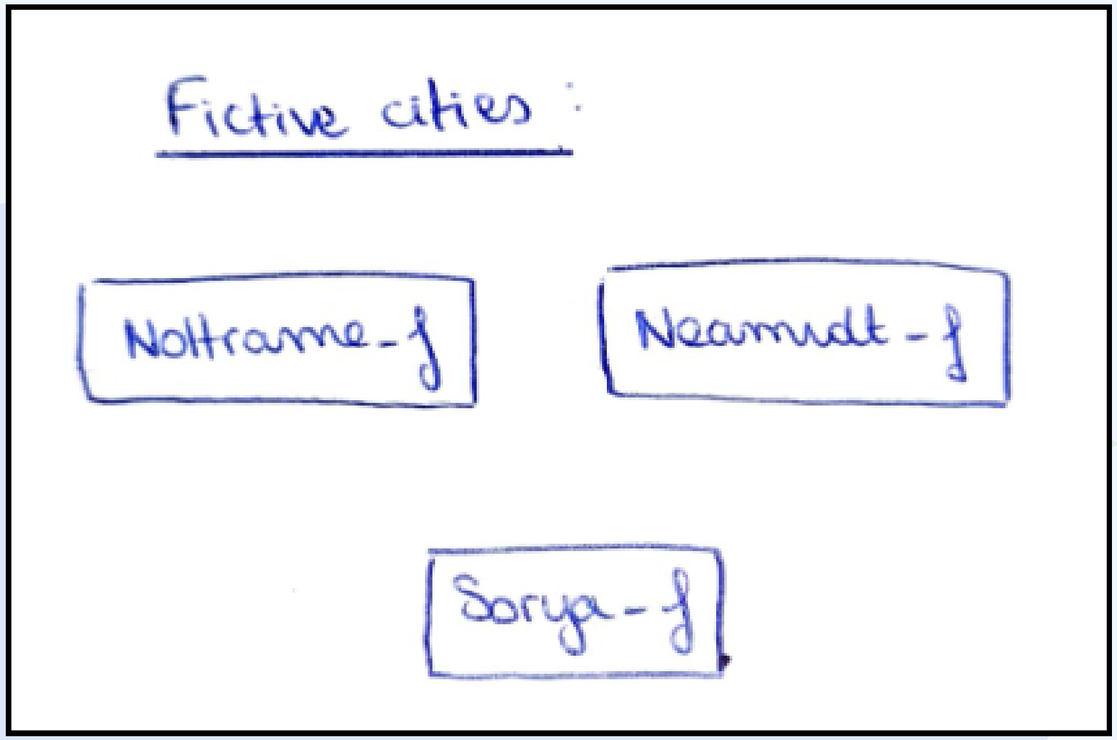
APPENDIX

WORK FLOW BETWEEN CITIES





The Noltrame_f is composed of all the people that wok in Noltrame.

=> no more flow between cities

FICTIVE CITIES EXPLANATION

A simplification of the work flow

Why introduce them?

Due to the work flow the PDE's of each city are coupled.
To de-couple them we introduced the fictive cities.

How ?

We created two tables , Workers and Resident :

- Workers[v][w] : the percentage of people working in w amongst the people that live in v
- Resident[v][w] : the percentage of people living in w amongst the people that live in w

We implemented a program that return (Sh_f, Ih_f,...) given (Sh, Ih,...) [code 1] and a program that returns (Sh, Ih,...) given (Sh_f, Ih_f,...) [code 2].

Then we resolve the linear system of fictive cities with Odeint and use the function above to have the results on the real cities.

FICTIVE CITIES EXPLANATION

Code 1

```python
def fictive_vers_real(valeurs_f):  # valeurs_f={ 'ne' : [  Sh_f , Ih_f, ..]} avec Sh_f=[Sh_f(0) , Sh_f(1) , Sh_f(2), ...]
    T=len(valeurs_f['s'][0])
    valeurs_r ={'no': [[0  for t in range(T)] for i in range(6)], 's' : [[0 for t in range(T)] for i in range(6)] ,'ne' : [[0 for t in range(T)] for i in range(6)]}
    for v in villes:
        for i in range(T):
            valeurs_r[v][0][i]= residents['ne'][v]*valeurs_f['ne'][0][i] + residents['no'][v]*valeurs_f['no'][0][i]+residents['s'][v]*valeurs_f['s'][0][i]
            valeurs_r[v][1][i]= residents['ne'][v]*valeurs_f['ne'][1][i] + residents['no'][v]*valeurs_f['no'][1][i]+residents['s'][v]*valeurs_f['s'][1][i]
            valeurs_r[v][2][i]= residents['ne'][v]*valeurs_f['ne'][2][i] + residents['no'][v]*valeurs_f['no'][2][i]+residents['s'][v]*valeurs_f['s'][2][i]
            valeurs_r[v][3][i]= residents['ne'][v]*valeurs_f['ne'][3][i] + residents['no'][v]*valeurs_f['no'][3][i]+residents['s'][v]*valeurs_f['s'][3][i]
        valeurs_r[v][4]=valeurs_f[v][4]
        valeurs_r[v][5] = valeurs_f[v][5]
    return valeurs_r   # valeurs_r={ 'ne' : [  Sh , Ih, ..]} avec Sh=[Sh(0) , Sh(1) , Sh(2), ...]
```

Code 2

```python
def reel_vers_fictive(valeurs_r,workers):  # valeurs_r={ 'ne' : [  Sh , Ih, ..]} avec Sh=[Sh(0) , Sh(1) , Sh(2), ...]
    valeurs_f = {'no': [0 for i in range(6)], 's': [0 for i in range(6)], 'ne': [0 for i in range(6)]}
    T= len(valeurs_r['s'][0])
    for v in villes:
        for i in range(T):
            valeurs_f[v][0][i]= workers['ne'][v]*  (parameter) valeurs_r: Any  ers['no'][v]*valeurs_r['no'][0][i] + workers['s'][v]*valeurs_r['s'][0][i]
            valeurs_f[v][1][i]= workers['ne'][v]*valeurs_r['ne'][1][i] + workers['no'][v]*valeurs_r['no'][1][i] + workers['s'][v]*valeurs_r['s'][1][i]
            valeurs_f[v][2][i]= workers['ne'][v]*valeurs_r['ne'][2][i] + workers['no'][v]*valeurs_r['no'][2][i] + workers['s'][v]*valeurs_r['s'][2][i]
            valeurs_f[v][3][i]= workers['ne'][v]*valeurs_r['ne'][3][i] + workers['no'][v]*valeurs_r['no'][3][i] + workers['s'][v]*valeurs_r['s'][3][i]
        valeurs_f[v][4]= valeurs_r[v][4]
        valeurs_f[v][5]= valeurs_r[v][5]
    return valeurs_f    # valeurs_f={ 'ne' : [  Sh_f , Ih_f, ..]} avec Sh_f=[Sh_f(0) , Sh_f(1) , Sh_f(2), ...]
```

FICTIVE CITIES EXPLANATION

[résolution EDP villes fictives]

```
def resolution_villes_fictives():
    predictions_fictives={}
    for v in villes:
        param_values=param_values_tab[v]
        Y0_ville_f=Y0_fictives_tab[v]
        valeur_fictive_v = integrate.odeint(deriv, Y0_ville_f, time_values, args=tuple(param_values))

        predictions_fictives[v]=creer_tableau_en_fonction_t(valeur_fictive_v )
        #tableau = [[Shv(0),Ihv(0)..],[Shv(1),Ihv(1) ...]...] =creer tableau en fonction t=> [Sh , Ih,..]   avec Sh=[Sh0 , Sh1, ...]

    return predictions_fictives
```

[résolution villes réelles]

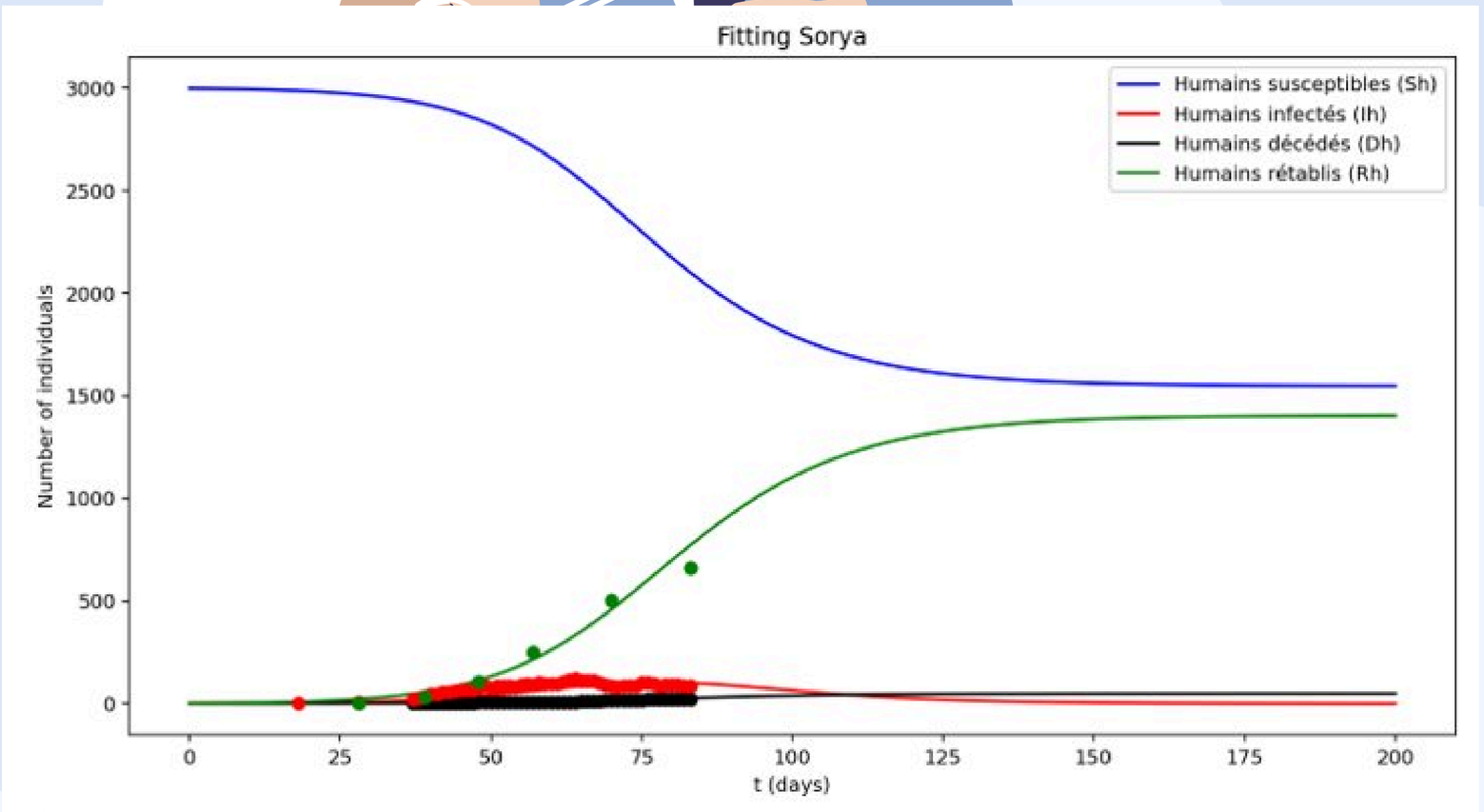
```
def resolution_villes_reeles():
    predictions_reeles= fictive_vers_real(resolution_villes_fictives())
    return predictions_reeles
```

OUR FIRST MODEL

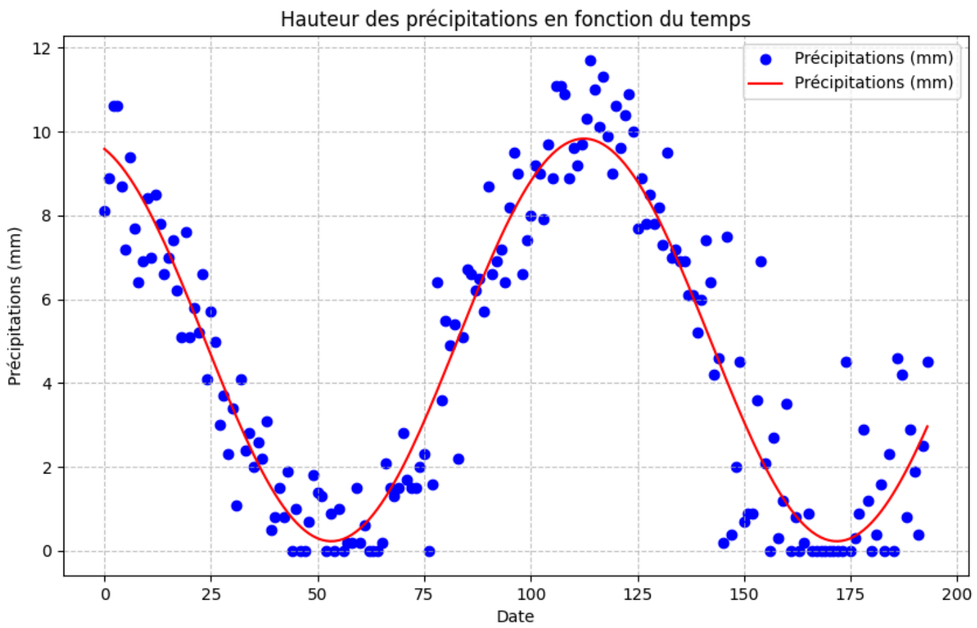
We started by studying a basic SIRD model with 6 different (constant) parameters to determine:

- Mosquito birth rate λ
- Mosquito death rate μ
- Transmission rates βh and βm
- Infection death rate ρ
- Recovery rate γ

This model allows for a basic fits that still manages to model Sorya's and Noltrame's evolution.



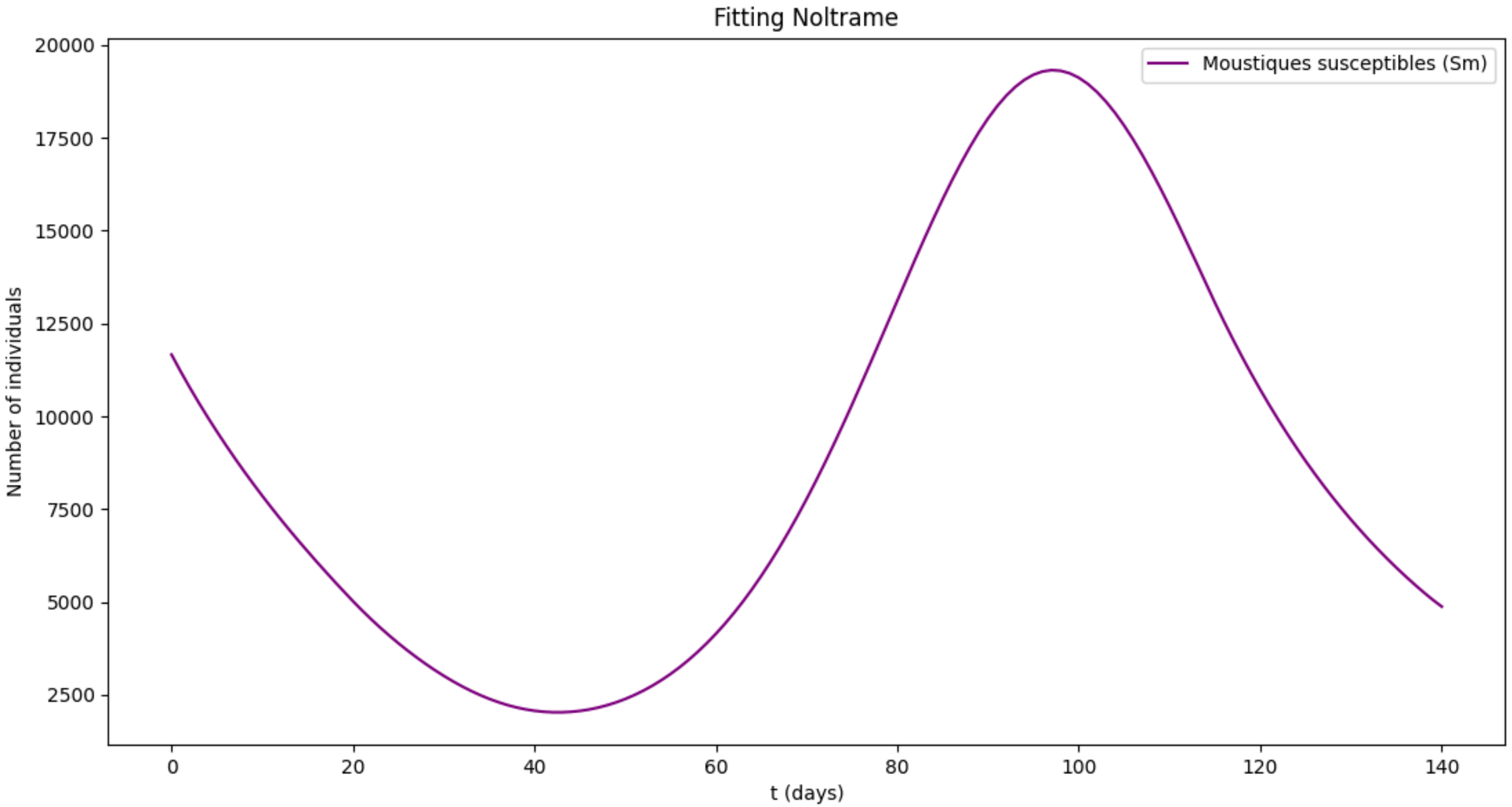
Precipitations



However, we realised that mosquito population did not evolve at a constant birth rate, but rather at a periodic one since they thrive in the humid season. So we adapted our model

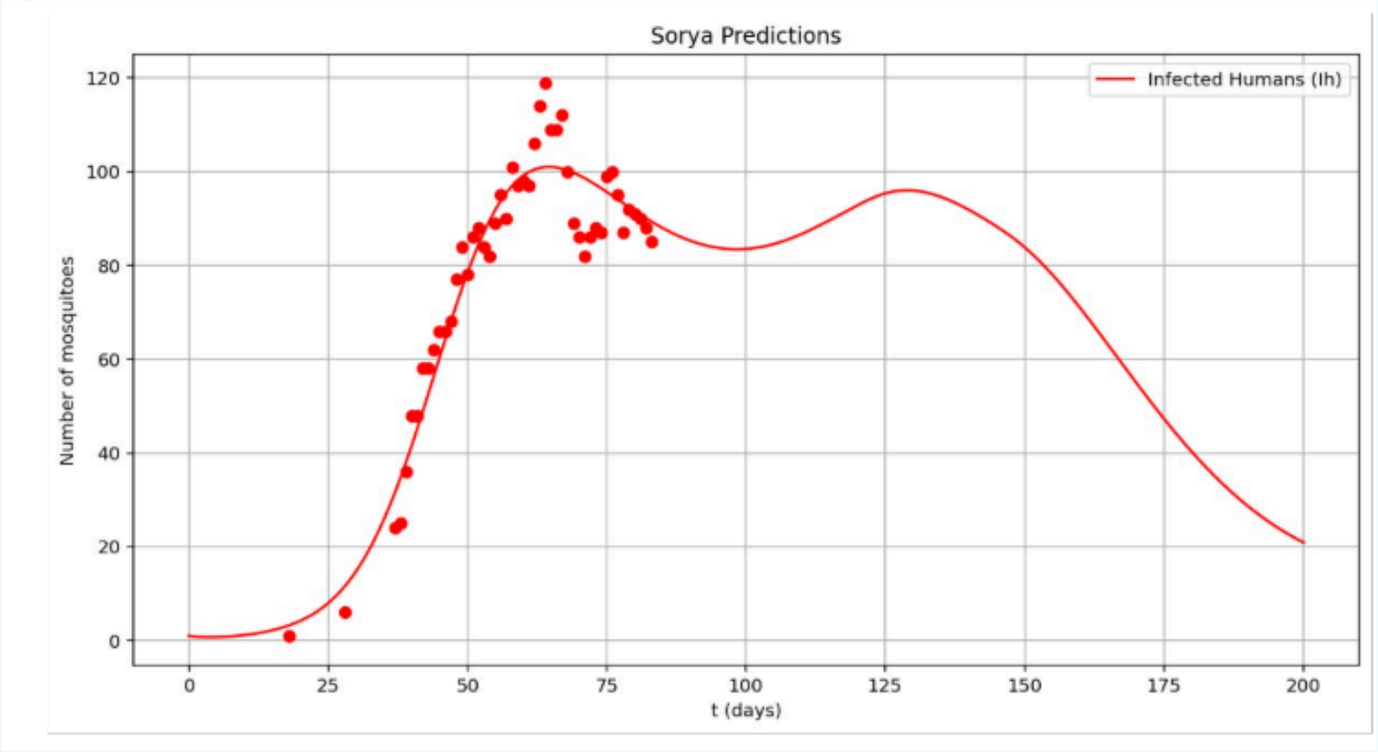
```python
def precipitation(t):
    x = 4.80*np.sin(0.053*(t+165)-4.39) + 5.03
    return x

def reproduction_rate(t):
    x = precipitation(t) + precipitation(t-1) + precipitation(t-2)
    y = 0.003181208*x - 0.01115436
    if y <= 0:
        return 0
    else:
        return y
```

Precipitations

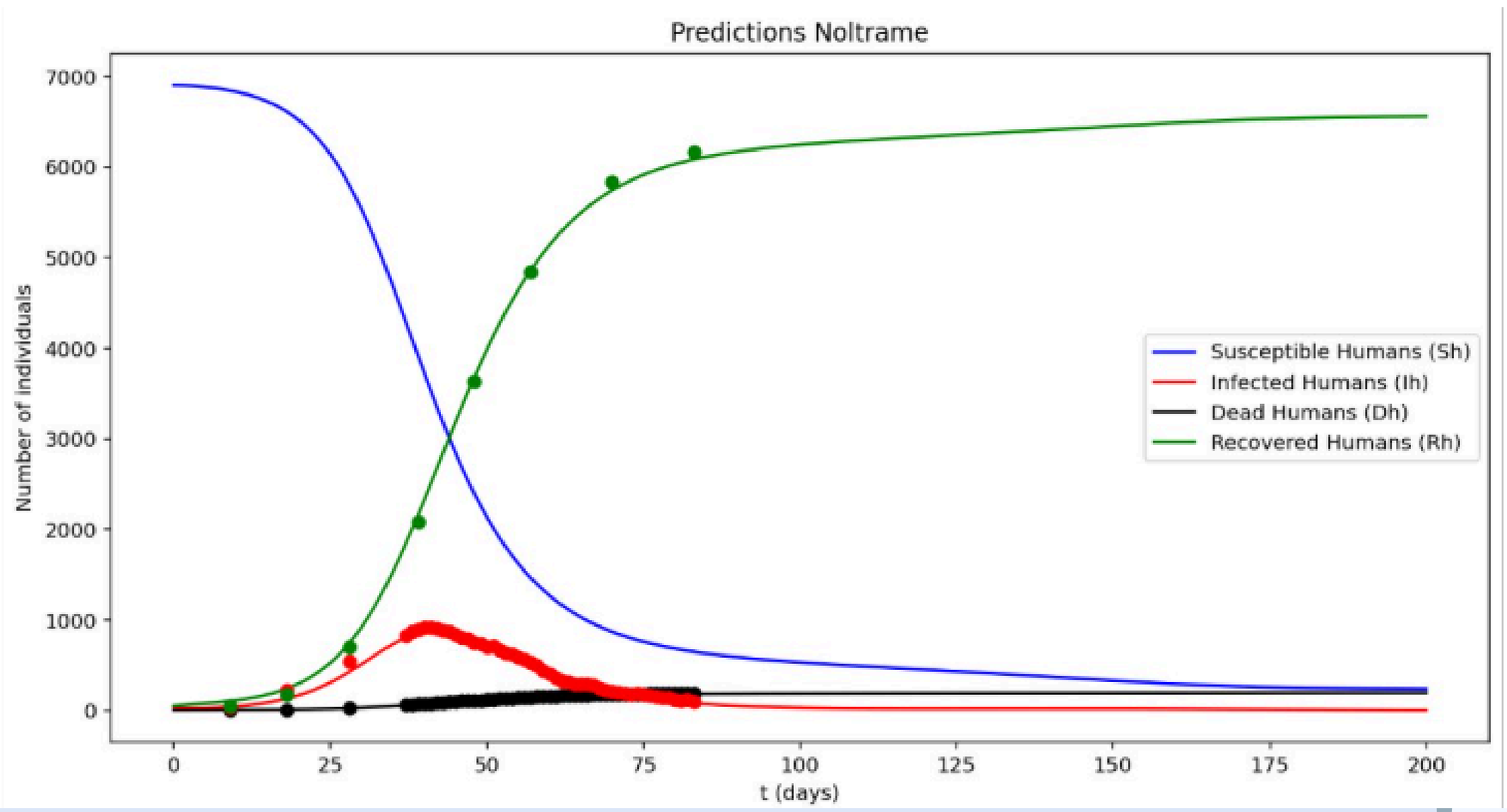


OUR FIRST MODEL

Taking into account mosquito reproduction rate gives the evolution of the infection a periodic behaviour as humid and dry season alternate.

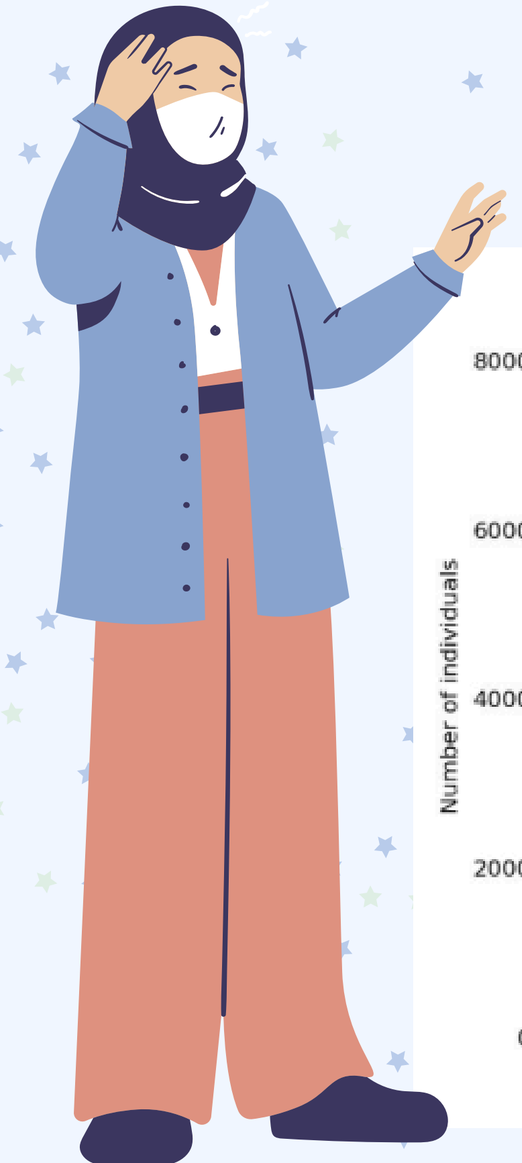


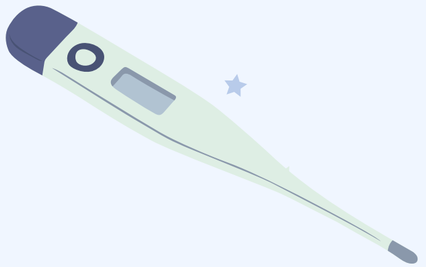
OUR FIRST MODEL

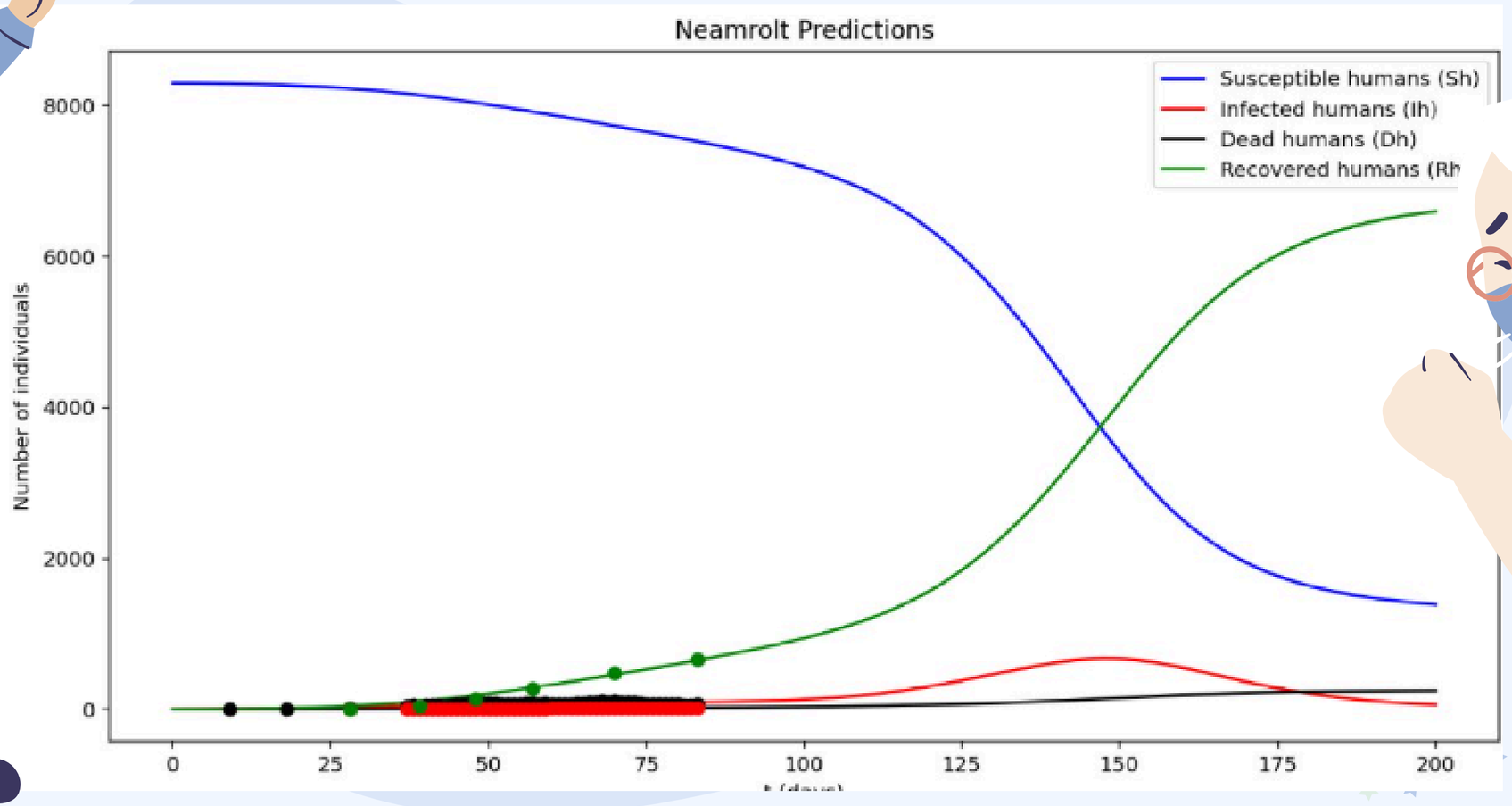


OUR SECOND MODEL

Considering human movement allows us to model Neamrolt, who has no mosquitoes but a lot of workers.







DATA TREATMENT

Data was collected in a SQL database

It was useful to get:

- The number of recovered in each city at different dates
- An approximation of some parameters like the duration of the epidemic.
- Some characteristics of the epidemic. It doesn't seem to depend too much on sex, no recovered can get reinfected

The detailed SQL code with explanations and the database are attached

SQL 1

```
1  SELECT * from data_utile where current_state like 'Dead %'
2  and hometown = 'Noltrame'
```

	jour	hometown	current_state	number
1	April_1	Noltrame	Dead / Hospitalized	5
2	April_10	Noltrame	Dead / Hospitalized	9
3	April_20	Noltrame	Dead / Hospitalized	25
4	May_1	Noltrame	Dead / Hospitalized	74
5	May_10	Noltrame	Dead / Hospitalized	110
6	May_19	Noltrame	Dead / Hospitalized	147
7	June_1	Noltrame	Dead / Hospitalized	178

Execution finished without errors.

SQL 1

```
1  SELECT count(*) from data_complete as d1 join data_complete as d2 on
2   d1.id = d2.id and d1.jour = 'April_20' and d2.jour = 'May_19' AND
3   d1.current_state = "Recovered" and d2.current_state like 'Infec%'
4
5   --Les 'Recovered' semblent ne pas retomber malades
```

	count(*)
1	0

FINAL MODEL

We have taken into account a paramater t and p which represent the time spent in a city (since they could work in a city and live in another

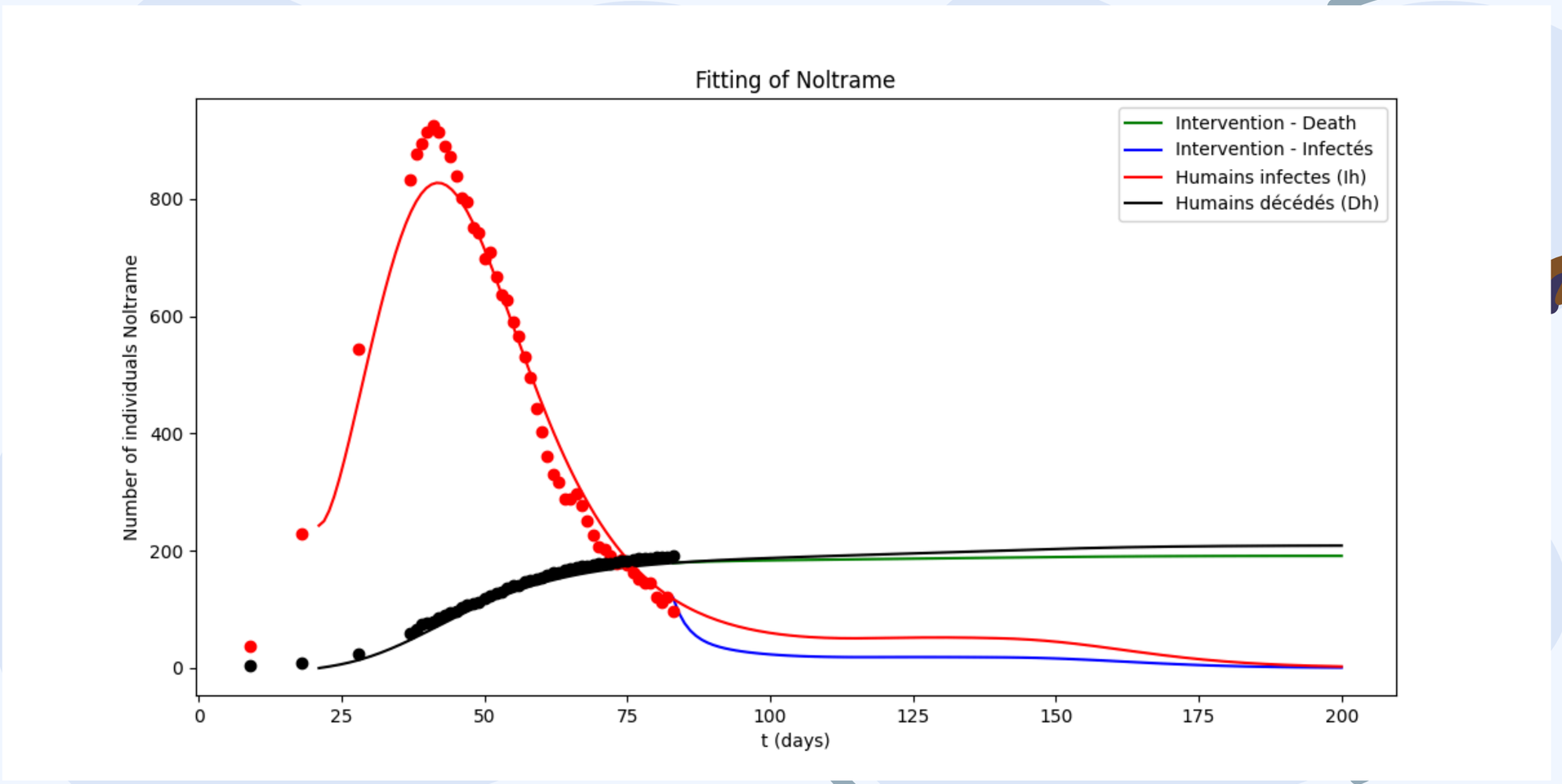
So they can get infected in the city they work or live in

We have a SIRD for each compartment of workers

We have a SI for each population of mosquitoes

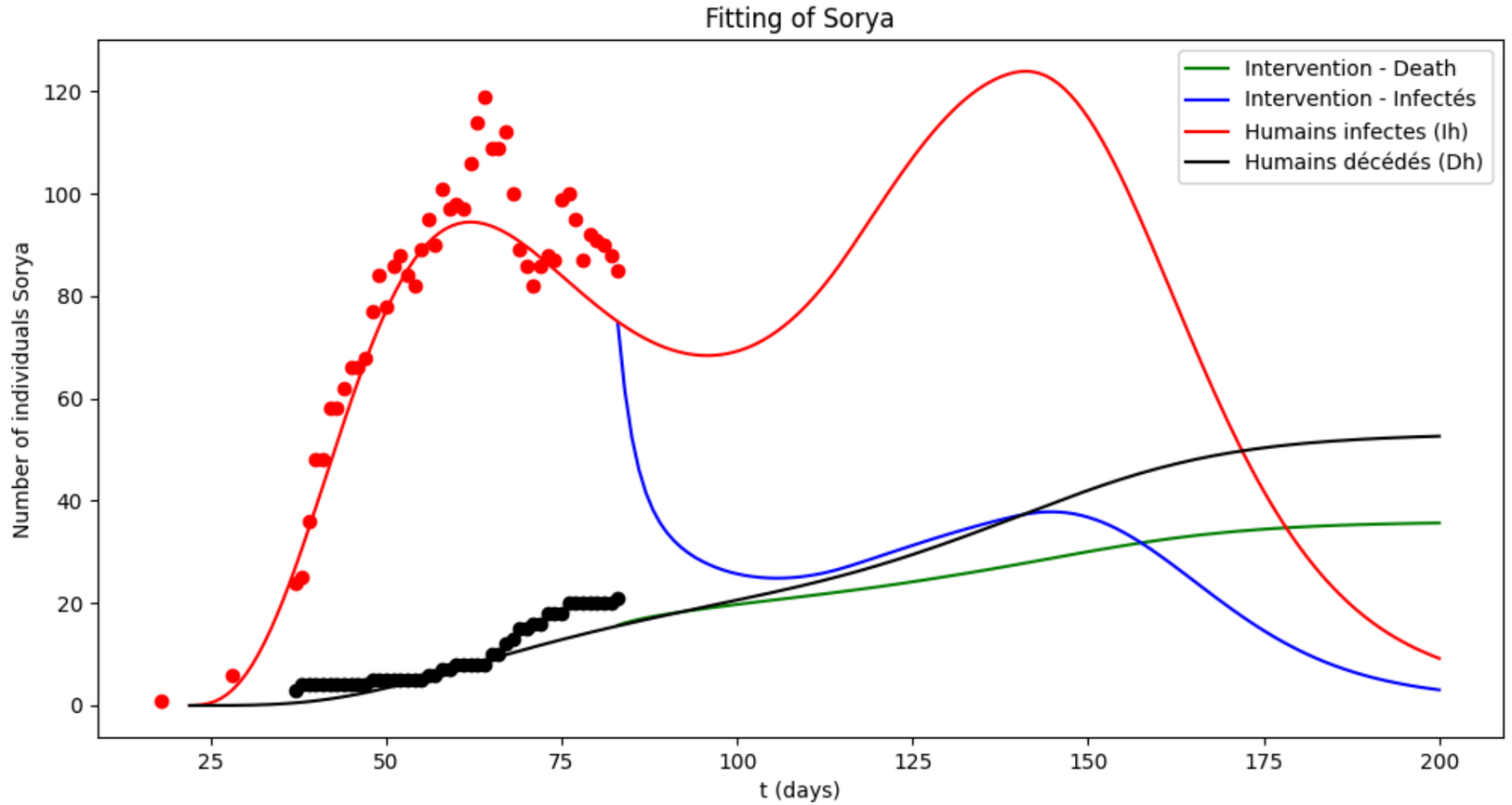
INTERVENTION

Standard treatment



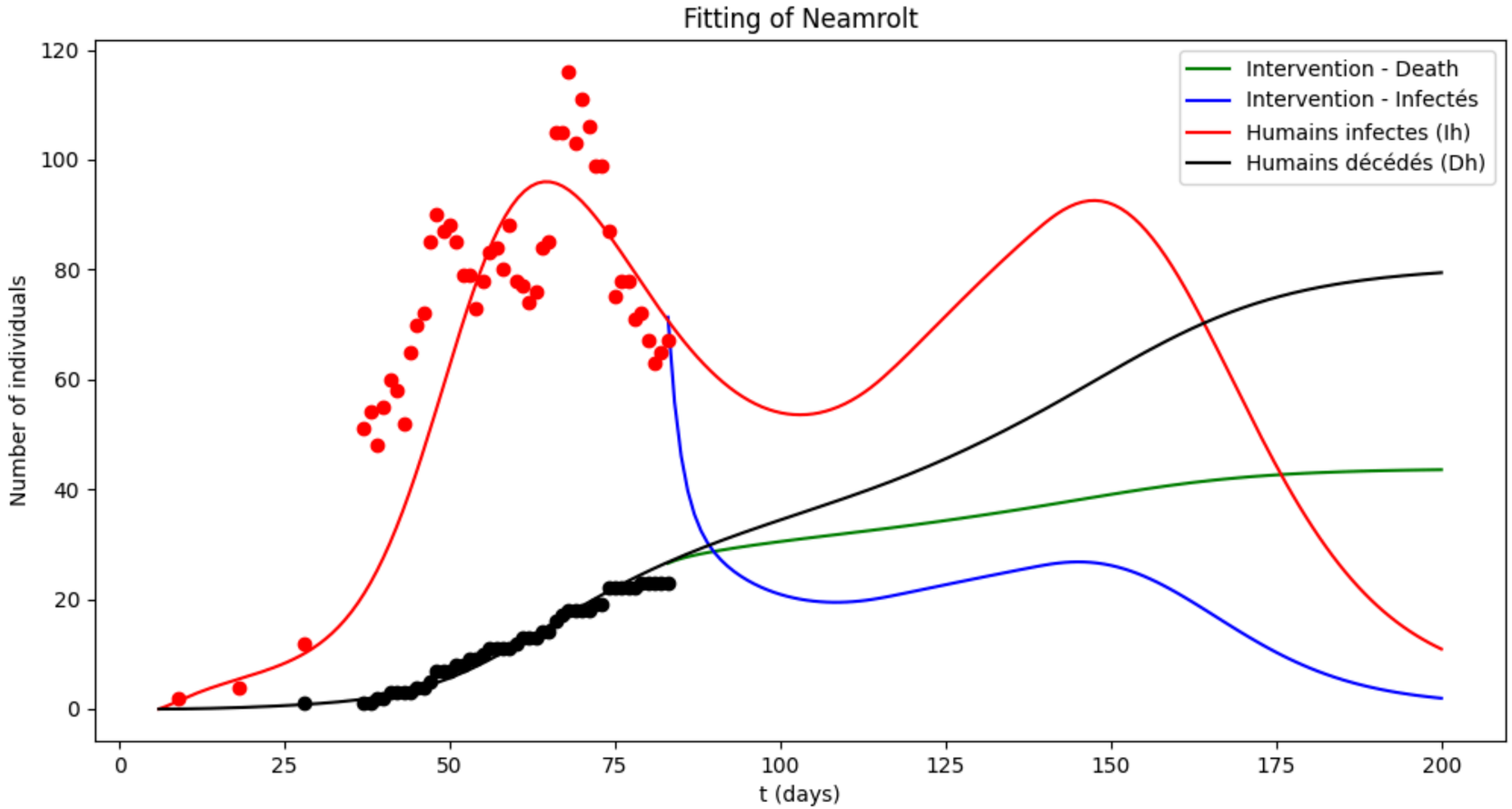
INTERVENTION

Standard treatment



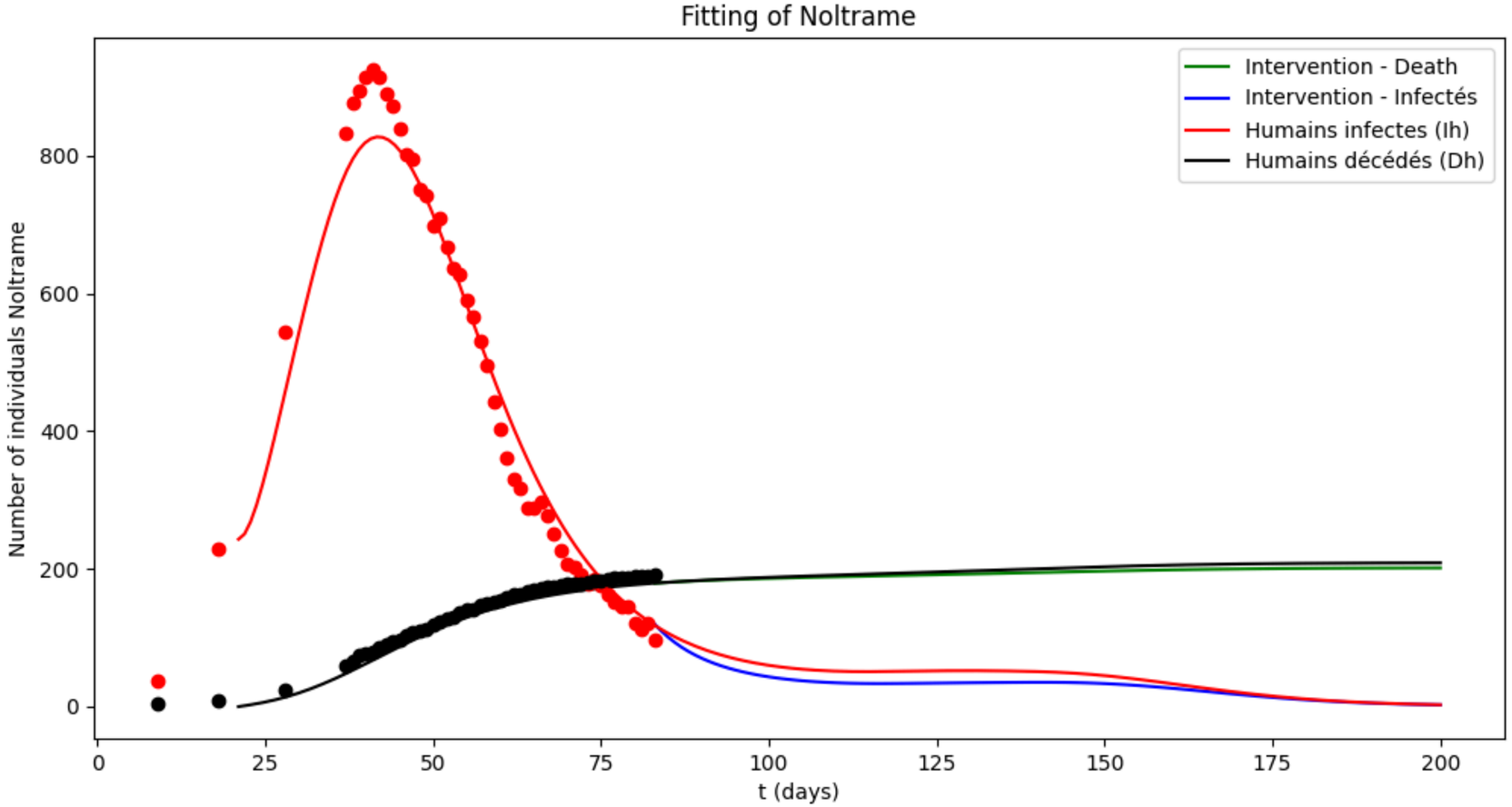
INTERVENTION

Standard treatment



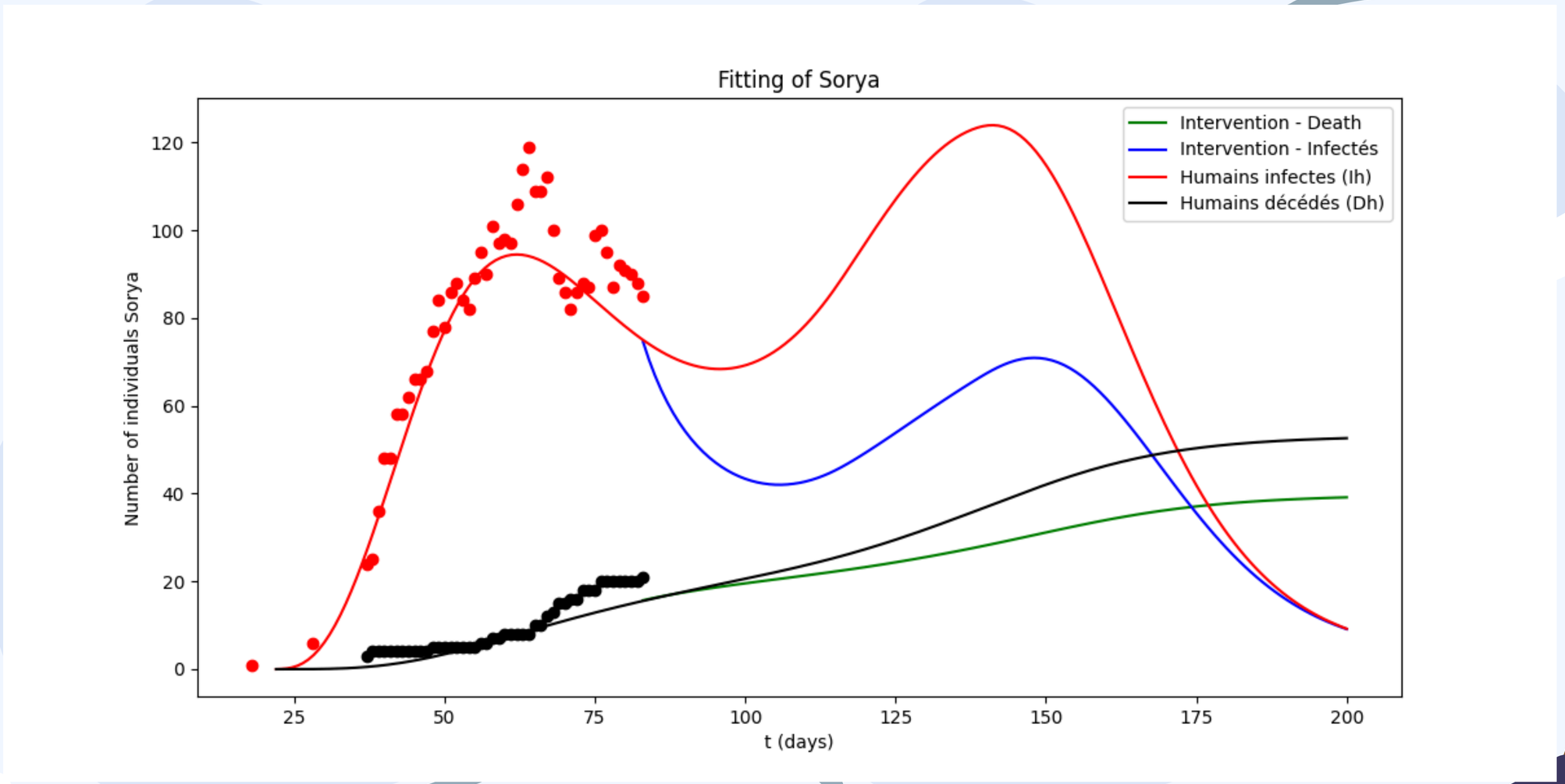
INTERVENTION

Spray (all inhabitants)



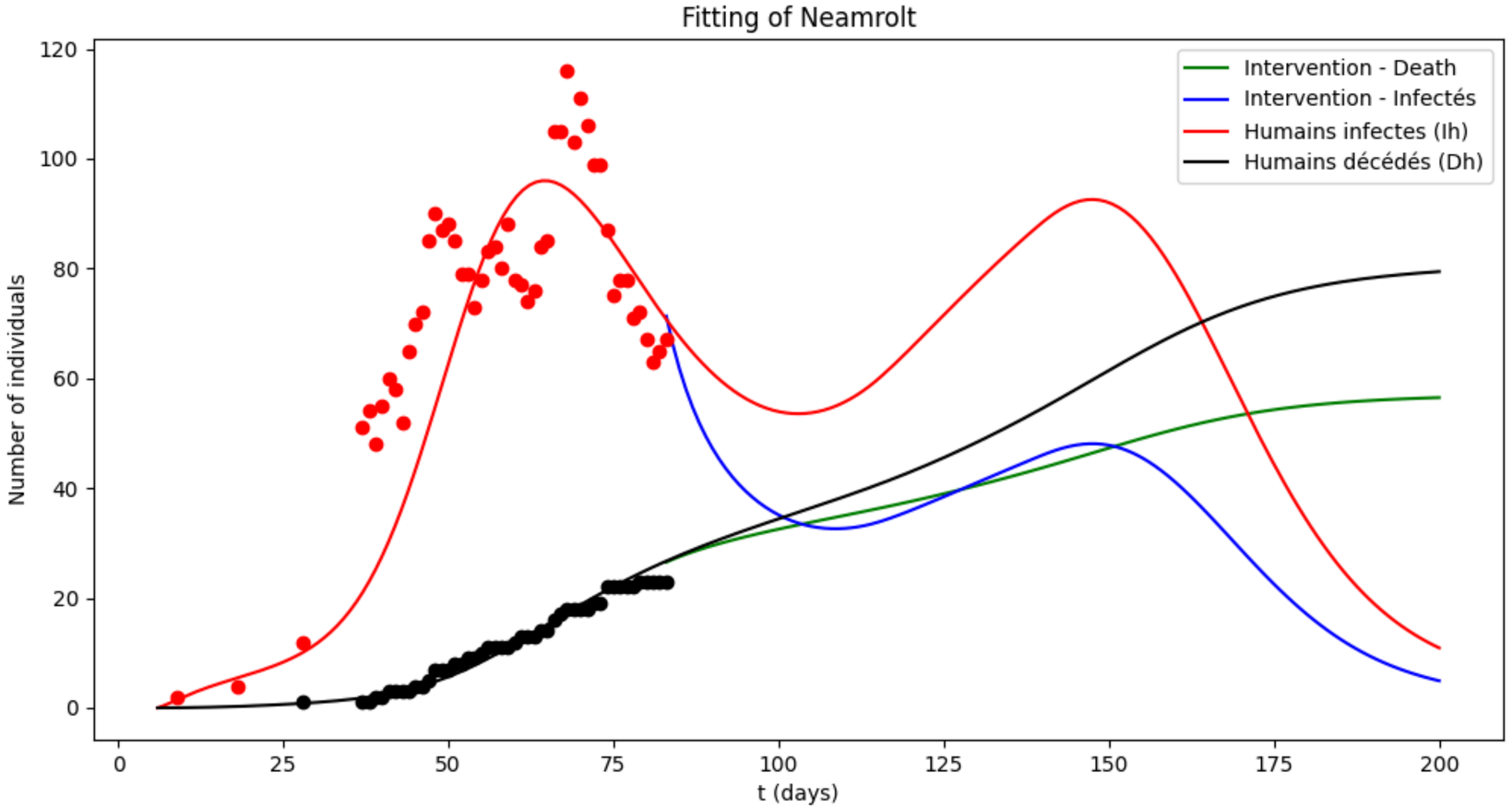
INTERVENTION

Spray (all inhabitants)



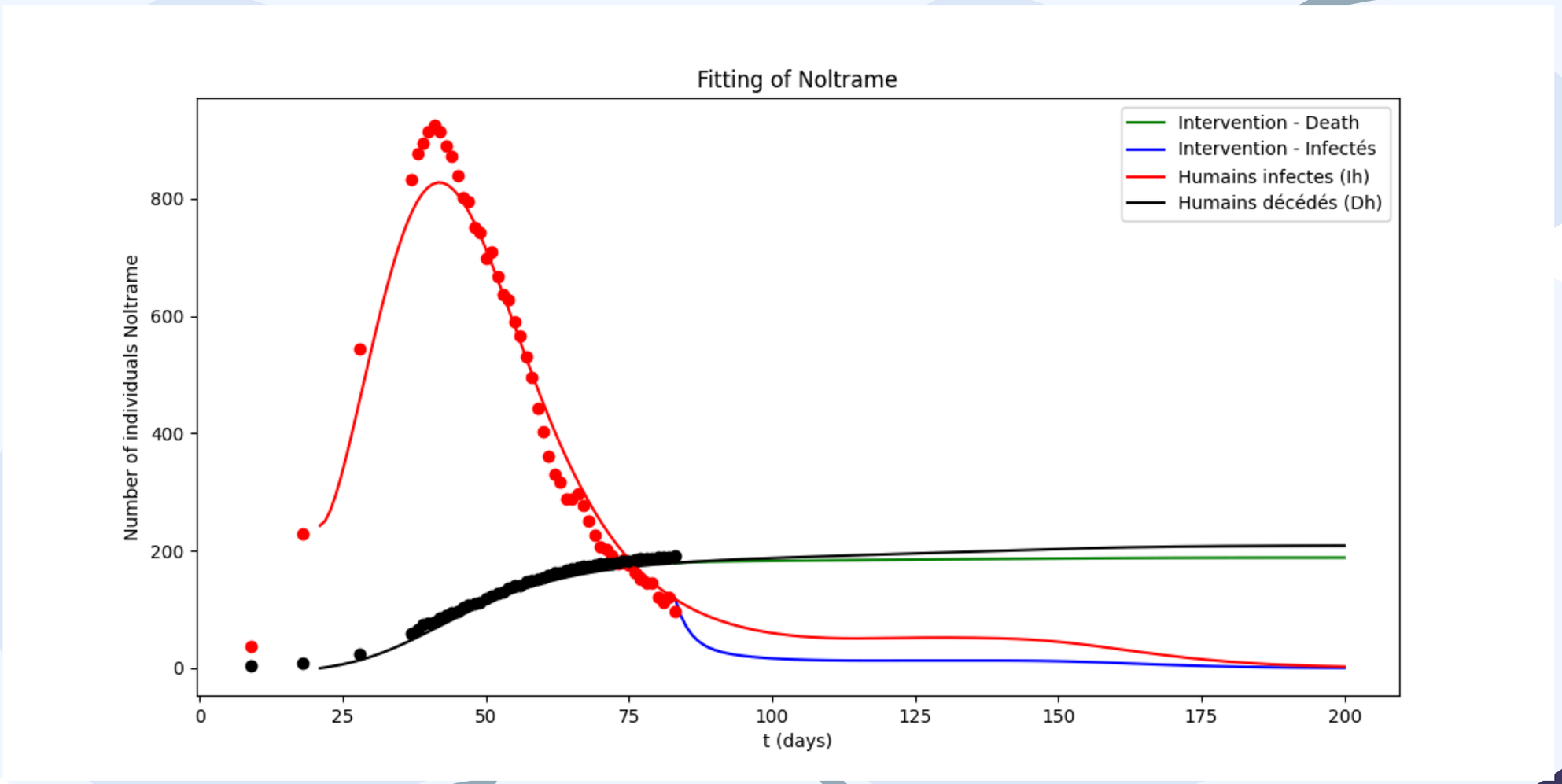
INTERVENTION

Spray (all inhabitants)



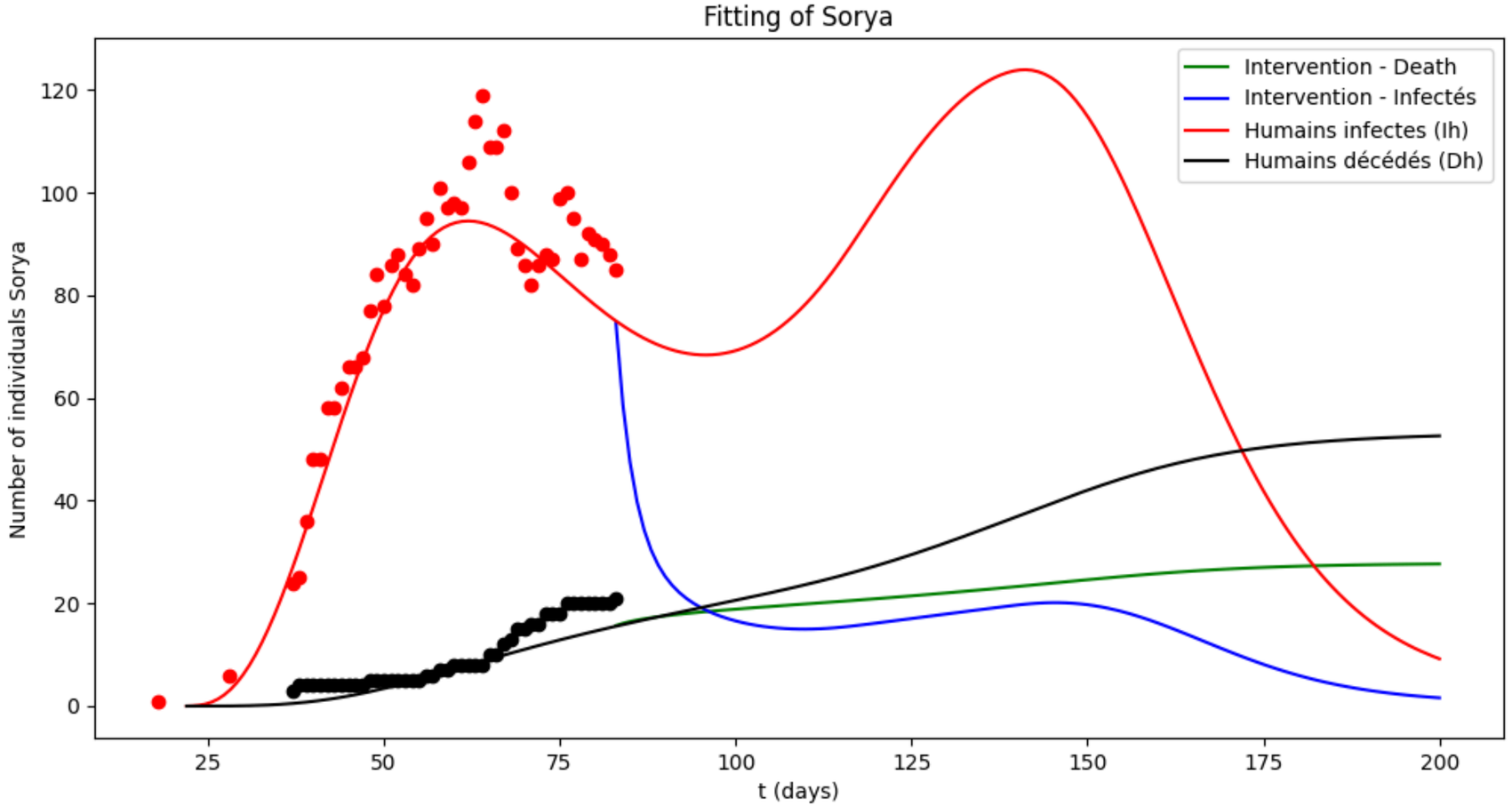
INTERVENTION

Full Intervention : Spray + treatment



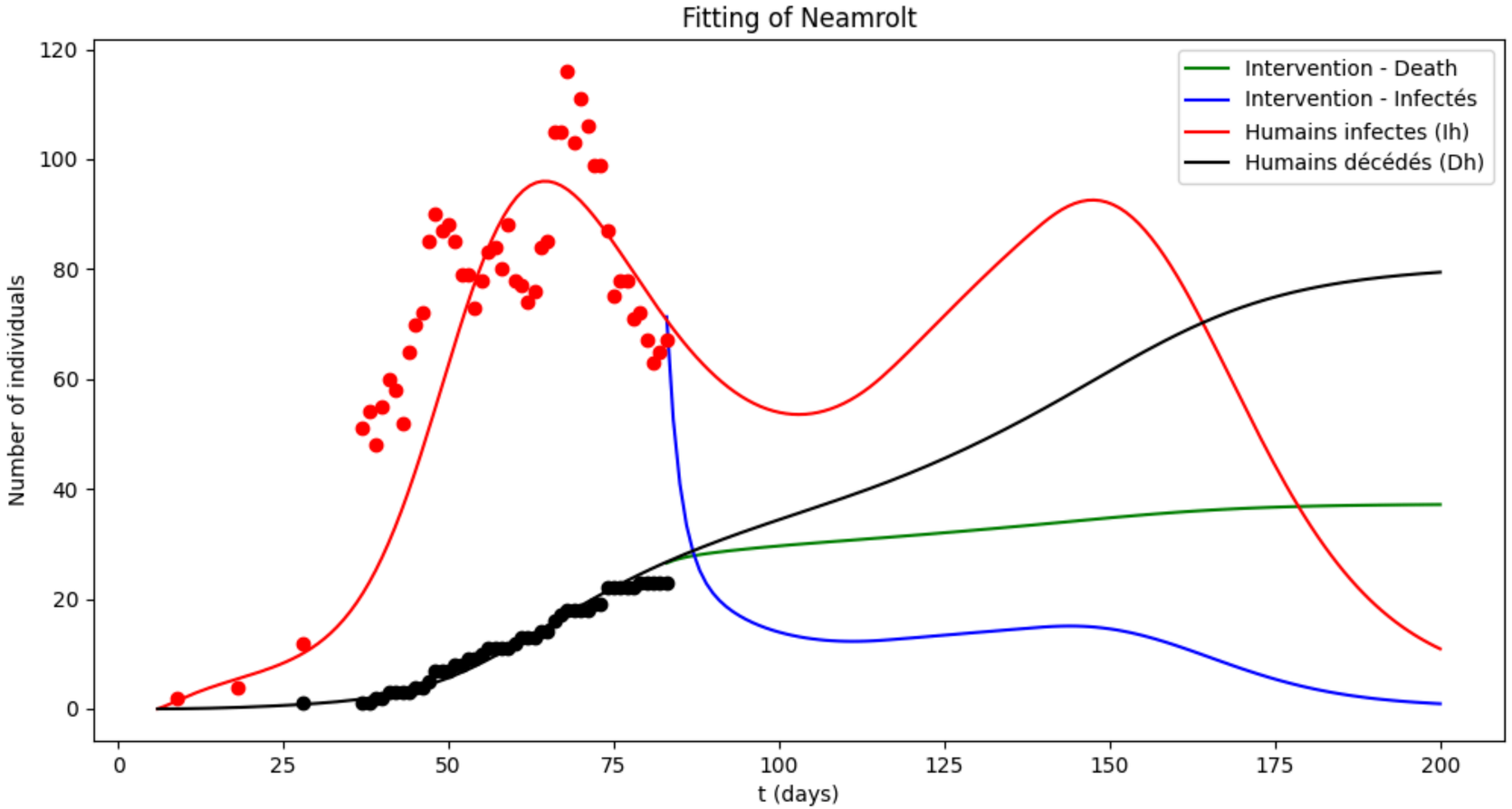
INTERVENTION

Full Intervention : Spray + treatment



INTERVENTION

Full Intervention : Spray + treatment



ORGANISATION

Communication :

- Within the team : we struggled to communicate well the 2 first days, but seeing that it slowed us, we did multiple little meetings to agree on plans of action for the next few hours
- With you : we often asked you question, but they weren't always spot on

Division of the work

- At first we were all working on our model but then we realised that we were wasting a lot of time. Then we decided to always have 2 to 3 people working on our model while others did other tasks like extracting data from the documents, SQL, the press release, slides, etc.

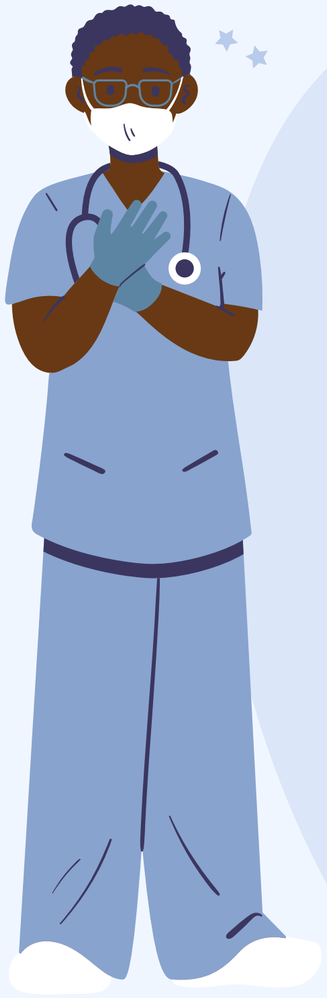
Sharing documents/code :

- WhatsApp, e-mails, Google Colab, Canva, Shared ppt

We should have done differently :

- Better organisation to code
- Better communication with the experts around us

Future Trends in Medicine



Predicting upcoming trends in medical science

Discussion on emerging technologies and research areas

Thank you for the challenge